Sun, Sand, and Services: Tourism and Household Welfare in Jamaica*

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Abstract

Tourism services have seen marked growth over the previous two decades. A number of lower- and middle-income countries have sought to take advantage of this boom in demand in the hopes of driving economic development. Even so, there remain significant questions about the ability of tourism to contribute to robust and inclusive prosperity for local populations in developing countries. I address this gap in the literature by investigating these questions in Jamaica, an upper-middle-income country that has made tourism the foundation of its development strategy. I combine an extremely rich and granular dataset of tourist expenditure surveys from the Jamaican Ministry of Tourism with a detailed, nationally representative household expenditure survey, both spanning nearly two decades. Linking tourist activity and individual households across consistent spatial units, I employ a shift-share instrumental variable identification strategy to estimate the effects of changes in tourism revenues on the welfare of local households. I find increases in real consumption and welfare for urban households with members working in mid-skilled occupations in non-tourism services and manufacturing. I discuss the implications of these findings for understanding the welfare impacts of tourism specialization in developing countries, and for the design of policies that aim to harness and scale tourism in the service of development.

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1 Introduction

The growth of service sector industries has become a defining feature of structural change in modern developing economies. Tourism has been one of the most important sources of growth within global services (World Tourism Organization 2023), and this has been especially true for lower- and middle-income countries (LMICs) (Nayyar et al. 2021). Despite the growth in tourism, there remain significant questions about its ability to generate economic growth that raises real consumption levels across a broad cross-section of the population. There is also uncertainty about the scope for tourism to generate positive spillovers to other sectors of the economy through forward and backward linkages.

Some research has found that regions specializing in tourism experience long-run economic benefits, in part driven by spillovers to manufacturing (Faber and Gaubert 2019). Other work has focused on the effects of urban tourism in large cities; these studies have found that the benefits to local residents in terms of income or improved amenities must be weighed against increases in the cost of living when determining welfare outcomes (Almagro and Domínguez-Iino 2025; Allen et al. 2021). It is also necessary to consider the extent to which the amenity preferences of locals align with those of tourists (Almagro and Domínguez-Iino 2025). Much of the existing work either has taken place in industrialized countries with large domestic tourism industries, or has focused on aggregate local and national effects rather than on household heterogeneity. Thus, there is a lack of evidence from developing economies on the short- to medium-run effects of tourism on individuals and households across different skill levels, industries, and incomes. This is partially due to data availability, and partially to the difficulty of measuring tourism activity. This paper contributes to the literature by investigating the relationship between tourism and household welfare, quantifying its effects on different skill segments of the labor force, and characterizing the heterogeneity of its impacts across socioeconomic segments. The study context is Jamaica, an upper-middle-income country that has made tourism a centerpiece of its development strategy.

I address two central questions. First, do increases in tourism activity in Jamaican municipalities increase real per-capita consumption expenditures for households within that municipality? Second, conditional on observing increases in real per-capita consumption, to what extent are these consumption increases observed among households below or near the poverty line? More broadly, I explore whether it is possible to leverage tourism for economic development that benefits the poorest in a community. I show that, in the short to medium run, increases in an area's tourism revenues increase real consumption levels, reduce poverty, and primarily benefit those in the lower quartile of the consumption distribution. These effects, however, are only present for urban households.

This study is motivated by the growth of Jamaican tourism, the heterogeneous exposure of individual communities across the island to the industry, and sizable year-to-year variation in performance of the sector. We can view booms in tourist activity in an area as demand shocks for local tourism services. Tourism is a labor-intensive sector; as a result, an increase in tourism can be expected to produce an increase in demand for local labor.

I harness two unique and granular datasets spanning 2001-2021: a detailed survey of tourist spending behavior and nationally representative household expenditure surveys.[1] To account for the likely endogeneity of local tourism, I employ a shift-share instrumental variable strategy. I exploit exogenous variation in the number of tourists from different regions of origin who choose to visit Jamaica, as well as variation in where in Jamaica they stay. There was significant temporal and spatial variation in tourism activity and household welfare across Jamaica during the period 2001-2021. Those years saw marked growth in the industry, but also considerable volatility; this variation enables me to identify the effects of tourism growth on the welfare of local residents.

A common criticism of tourism in Jamaica and other developing countries is that it is an exploitative industry that has little connection with the local populations (Çiftçi et al. 2007; Issa and Jayawardena 2003; Issa and Jayawardena 2003). Debates about tourism have drawn comparisons to natural resource ventures (Ngassam et al. 2024). Both situations involve tradable goods or services — that is, demand for a good or service by non-local people. To guide this analysis, I draw not only from research on tourism (Allen et al. 2021; Almagro and Domínguez-Iino 2025), but also on studies of local demand shocks and resource booms (Moretti 2010; Moretti 2011; Aragón and Rud 2013). The framework developed by Moretti (2010) and Moretti (2011) to evaluate local labor demand shocks predicts that an increase in demand for labor-intensive services such as tourism will raise local earnings (and therefore local consumption levels), relative to earnings in communities that do not experience the shock. This may then result in higher earnings for workers in other sectors because of higher consumption levels, owing to larger local budget constraints. However, a positive tourism shock may also produce inflationary effects because of the increase in local earnings (Moretti 2010; Aragón and Rud 2013), or because of more tourists consuming local goods and services (Allen et al. 2021; Almagro and Domínguez-Iino 2025). Aragón and Rud (2013) predict that the skill levels of the workers who benefit will depend on the skill composition of the industry experiencing the demand shock. Given that lower- to mid-skilled occupations comprise 83 percent of the Jamaican tourism workforce[2], it is reasonable to expect a tourism shock to yield gains for those workers. Finally, because those in the lower segments of the expenditure distribution are more likely to work in low- to mid-skilled occupations,[3] I would expect households below or near the poverty line to experience the greatest benefit in the Jamaican setting.

I find that greater tourism revenues in a development area have a positive effect on real per-capita expenditures.[4] In particular, a difference in tourism revenues of 10 million US Dollars (USD) within a development area generates a 2.7 percent increase in per-capita expenditures for urban households relative to households

[1] In developing countries, real household expenditures or consumption often provide a more accurate representation of household welfare than income-based measures. Consumption tends to be more accurately reported than income (Carletto et al. 2022), is less influenced by short-term fluctuations in earnings, and better captures differences in assets and access to credit (Meyer and Sullivan 2003).

[2] Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions.

[3] 91 percent of households in the lowest quartile of the expenditure distribution are employed in low- or medium-skilled occupations, versus 69 percent of households in the highest quartile. Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions.

[4] "Development areas" are divisions used by the Jamaican Social Development Commission and the Statistical Institute of Jamaica to group communities based on geographic, economic, social, and demographic characteristics.

in other development areas.[5] Because the median annual difference in tourism revenues among development areas is 7.9 million USD, this implies that households in the median development area have 97 USD higher real per-capita expenditure. Households increase their food and non-food expenditures by 2.3 percent and 2.9 percent, respectively, although these are not statistically different from one another. While within the non-food expenditure category there are no statistically significant effects on education or loan repayment, I do find a statistically significant increase in spending on medical goods and services. All told, increased real expenditures on medical services and products represent 33 percent of the change in non-food expenditures. Lending further support to the interpretation of these effects as improvements in well-being, households are .6 percent less likely to be below the poverty line for every 10 million USD increase in tourism revenues.

As expected, there is an increased likelihood that a surveyed household works in a tourism-related industry following a positive shock to development area tourism revenues.[6] Surprisingly, the increases in real expenditures are driven by those who work in non-tourism sectors, who on average see an increase in real per-capita spending of 2.9 percent per additional 10 million USD in tourism revenues annually. There is no statistically significant effect on real expenditures for households working in the tourism sector. Similarly, decreases in poverty are entirely driven by non-tourism households. I subsequently show that it is urban mid-skilled occupations (skill level 2) that are the primary beneficiaries of increased tourism revenues.[7] The level 2 occupations that benefit are "Service, Craft, and Trade Work" and "Plant and Machine Operators". Using a quantile regression, I find that urban households in the bottom two quartiles of the urban consumption distribution benefit from increased tourism levels.

Finally, I provide additional support for these results by utilizing an unbalanced panel of households that is a subset of the main repeated cross-section. The panel regressions show that the effects of increases in tourism revenues are concentrated among the households that are employed in non-tourism services and manufacturing during all periods in which they are observed. The households that benefit on average are those whose per-capita expenditure falls into quartiles 1-3 of the consumption distribution (with quartile 1 the lowest) in the first period in which they are observed. In order to confirm that these results are not driven by cost of living increases, I run two additional regressions. The first shows that there is no effect of tourism earnings on real rent expenditures for households. The second shows that the effects of tourism are driven by workers in private sector employment, with no change in real consumption for government workers, whose salaries are set nationally, and are unlikely to reflect changes in local tourism levels. These results further support the argument that the observed increases in expenditures reflect increases in real consumption and welfare.

Jamaica is an informative setting in which to study the relationship between tourism and local household

[5] Urban households have an average per-capita annual expenditure of 3990.83 USD per-capita as shown in table 2

[6] Only the employment of the principal earner or the head of household is considered in these calculations. Going forward, whenever employment or work of a "household" is mentioned, this is referring to the employment of the principal earning member of the household or the head of household.

[7] This classification is based on the 2008 International Standard For the Classification of Occupation (International Labour Office 2012) developed by the International Labour Office. Level 1 (elementary) occupations require very little or no training or schooling. Level 2 occupations generally require up to a secondary education. Level 3 and 4 occupations comprise professionals, technicians and managers, and typically require some kind of tertiary-level training. I will refer to Level 1 occupations as Low-Skilled or Unskilled occupations, level 2 as Mid-Skilled, and levels 3 and 4 as High-Skilled.

welfare for two reasons. The first is that Jamaica is comparable to a key subset of countries in its level of economic development and industrialization, the type of tourism in which it has a comparative advantage, and the skill levels of its labor force. Jamaica is a middle-income country heavily dependent on services, which make up 70 percent of its economic output. This is similar to the large group of developing economies increasingly specializing in services and failing to industrialize (Rodrik 2016). An island nation located in the Caribbean Sea and endowed with world-famous beaches and a tropical climate, Jamaica has specialized in the "Sun, Sand, and Sea" tourism that is popular in such climates. Regions endowed with similar climates include Latin America, Sub-Saharan Africa, Southeast Asia, and the South Pacific.

As is the case for many middle-income countries, Jamaica's labor force is overwhelmingly made up of workers qualified for low- and mid-skilled occupations. A primary objective for emerging economies is to develop industries that are able to absorb this large supply of labor; this is one of the major strengths of tourism (Nayyar et al. 2021). The second attribute of Jamaica that makes it useful for study is the country's aggressive pursuit of tourism-led growth. Jamaica's shared characteristics with a number of countries and its pursuit of tourism-oriented development mean that findings from this setting can provide insights relevant to a number of other countries. These findings may be useful for countries that have also pursued tourism-led growth and wish to better understand their outcomes, or they can be informative for places considering such a growth strategy. The interest in tourism as an engine of development can be seen in the diverse set of countries that have developed strategic plans for expanding their tourism industries. These include Indonesia[8], Botswana[9], and India.[10]

This paper contributes most directly to the small but growing literature studying the effects of tourism on local and national economies. One segment of this literature has focuses on utilizing aggregated regional or country-level outcomes in order to quantify the impacts of tourism. In the context of Mexico, Faber and Gaubert (2019) investigate the long-run effects of tourism specialization and find that tourism-producing municipalities have experienced significant long-run gains relative to non-tourist regions. These outcomes are due in part to backward linkages to the manufacturing sector. In Thailand, Wattanakuljarus and Coxhead (2006) find that growth in tourism raises aggregate income, but worsens inequality. A study by Croes et al. (2021) in Poland finds that tourism specialization generates short-run economic gains, but these gains do not persist in the long run. Another emerging strand of the tourism literature has investigated the welfare impacts of tourism on local residents in urban centers, such as Barcelona (Allen et al. 2021) and Amsterdam (Almagro and Domínguez-Iino 2025). These studies focus on the "crowding out" effects of tourism on local residents in terms of living expenses or amenities. For Barcelona, Allen et al. (2021) show that tourism's effects on the well-being of local residents exhibits a tension between tourism-driven wage growth and tourism-induced increases in the cost of living. They find that, while local workers suffer on average, there is substantial heterogeneity in effects across space. In the case of Amsterdam, Almagro and Domínguez-Iino (2025) reveal that the aggregate welfare effects of more tourist-focused amenities on local residents are a function of the distribution of local preferences for amenities,

[8] Ministry of Public Works and Housing 2018.

[9] Botswanan Ministry of Environment, Natural Resources, Conservation, and Tourism 2021.

[10] Indian Ministry of Tourism 2022.

as well as resident demographic characteristics such as age. My paper is most similar to Allen et al. (2020) in my usage of a shift-share instrumental variable strategy exploiting variation in tourists' country of origin, and in the main goal of quantifying the welfare effects of tourism on local residents. My paper differs significantly, however, across a number of dimensions. These include the economic context in which the research takes place, the type of tourism that is dominant in this context, the time period covered by the analysis, and, more generally, the focus of my paper on investigating the capacity of tourism to drive broad economic development.

This study builds upon the existing literature on tourism in four specific ways. First, through harnessing the rich household data, this study is able to provide answers about how the earnings of tourism accrue differently to individual households across characteristics such as socioeconomic status, skill level, industry of employment, and geography. Furthermore, through analyzing disaggregated components of consumption, this paper is able to better capture key aspects of household-level well-being in a manner that is not possible in aggregated regional or national analyses.

Second, the temporal breadth of the data enables this analysis to capture the effects of significant and uneven growth of the tourism sector across space in Jamaica. This provides useful variation for accurately estimating the true effects of tourism on local residents.

The third way in which this paper adds to existing work on tourism is that it analyzes the effects of tourism on households for a type of tourism that is different in important ways from the type that takes place in some other studies. Whereas tourism in Amsterdam and Barcelona occurs primarily in residential areas and generally is well-integrated with the urban environments (Almagro and Domínguez-Iino 2025; Allen et al. 2020), Jamaican tourism is largely characterized by tourists choosing "all-inclusive" vacation packages.[11],[12] We may therefore expect the implications of tourism booms for spillovers to other sectors, or crowding-out effects on locals, to be different from those in integrated urban tourism settings. Furthermore, tourism services in Jamaica are overwhelmingly exported, with more than 90 percent of Jamaican tourism revenue coming from international visitors in 2023 (STATIN 2019), compared to approximately 20 percent in Mexico (Faber and Gaubert 2019). To the extent that the export share of tourism is associated with different skill requirements for workers, interactions with the local economy, and vulnerability to external shocks, this distinction could have important implications for the welfare effects of tourism on local residents.

Fourth, this paper investigates the impacts of tourism in an emerging economy that has yet to substantially industrialize,[13] and which is largely specialized in services.[14] Thus, we may expect that the scope for tourism to have spillovers to other sectors may differ from countries with substantial manufacturing activity, such as Mexico.[15]

[11] Jamaican Ministry of Tourism 2025.

[12] This "enclave" approach is common in developing country contexts (Issa and Jayawardena 2003) and is often characterized by visitors spending significantly less time and money in the communities in which they vacation (Çiftçi et al. 2007).

[13] The manufacturing value-added share of GDP in the Jamaican economy was roughly 8 percent in 2024 according to The World Bank Development Indicators: https://data.worldbank.org/indicator.

[14] Services comprised 60 percent of 2024 GDP for Jamaica, according to the World Bank Development Indicators.

[15] Manufacturing value-added comprised 20 percent of Mexican GDP in 2024, according to The World Bank Development Indicators:https://data.worldbank.org/indicator.

In addition to contributing to the tourism literature, this study also contributes to the literature at the intersection of service-led structural transformation and economic development. In recent years, there has been significant research into the declining relative share of manufacturing in many emerging economies, called "premature de-industrialization" by Rodrik (2016). This trend has raised concerns, because manufacturing growth has traditionally been the means by which countries improve their productivity, grow their economies, and raise living standards. There is still significant uncertainty about whether or not services such as tourism are capable of filling the productivity-enhancing role traditionally played by manufacturing.

Studies such as Nayyar et al. (2021) view services such as tourism as unlikely to share the ability of manufacturing to absorb low-skilled labor while also generating productivity growth. Some work, however, has shown the potential strengths of specializing in services. For example, Fan et al. (2023) show that India gained from service sector specialization, in large part because of the increase in productivity in urban consumer services that benefited wealthy urban dwellers. Research by McCullough (2025) in Tanzania shows that, while service-focused structural transformation has the ability to generate local employment, the scope for long-term growth may be limited due to the constrained size of local markets and low productivity growth. I provide insight into the implications of tradable service sector specialization for local residents, and also characterize the effects of shocks to a tradable service sector on heterogeneous households.

Other studies have detailed the potential challenges of structural change that is not driven by industrialization. Gollin et al. (2016) and Venables (2017) discuss how gains from urbanization depend on whether or not the industries in an urban area exhibit growth in productivity. They also show how urbanization that draws large numbers of people into lower- productivity non-tradable services could, in fact, dampen long-term growth. This type of urbanization could contribute to effects consistent with the "Dutch disease" framework developed by Corden and Neary (1982), or the "Cost Disease" framework introduced by Baumol and Bowen (1965). I provide empirical evidence of how shocks to a major tradable service sector impact household welfare in a country where the economy has progressed directly from being based upon natural resources to a focus on services.

Finally, this paper contributes to the literature studying the effects of sectoral shocks on local economies. A significant subset of this literature relates to the effects of resource windfalls or natural resource ventures on local communities (Aragón and Rud 2013; Aragón and Winkler 2023; Bonilla Mejía 2020; Boire 2021). Other studies have investigated the effects of manufacturing growth on decisions regarding human capital accumulation in settings such as Bangladesh (Heath and Mushfiq Mobarak 2015) and Mexico (Atkin 2016). I provide novel insights into the magnitude of the effects of tourism booms on the well-being of local households in a developing country. As a major global sector that has experienced considerable growth and has come to dominate many economies, tourism's effects on the communities it impacts is of great relevance.

In Section 2, I present background information on the Jamaican economy and tourism. Section 4 details the effects of tourism shocks predicted by the literature and lays out four testable hypotheses. Section 5 explains the datasets used for the study, while Section 6 presents the empirical specification. Section 7 explains the findings of the study. Section 8 discusses the implications of the findings and concludes.

Table 1: Global, Regional, and Jamaican Tourism Statistics

Indicator	Value
Panel A: Global & Caribbean Region Tourism Statistics	
International Tourist Arrivals (2023)	1.3 billion USD (UNWTO)
International Tourism Export Revenues (2023)	1.8 trillion USD (UNWTO)
Global tourism GDP share (2023)	3% (UNWTO)
Share of Global Trade in Services	23% (UNWTO)
Average Growth In International Arrivals (2000-2023)	4.1% annually (UNWTO)
Caribbean Tourism arrivals (2024)	34.2 Million (CHTA)
Tourism Average Share of Caribbean GDP (2015-2019)	25.4% (OECD)
Panel B: Jamaican & Caribbean Tourism	
Total Tourist Arrivals to Jamaica (2024)	4.15 Million (MOT)
Total Stopover Arrivals to Jamaica (2024)	2.9 Million (MOT)
Total Cruise Passenger Arrivals to Jamaica (2024)	1.25 Million (MOT)
Total Roomnights Sold in Jamaica (2024)	5.75 Million (MOT)
Jamaica Total Tourism Earnings (2024)	4.2 billion USD (MOT)
Panel C: Jamaican Tourism Sector Capital Stock & Infrastructure	
Jamaica Hotel Room Capacity (2024)	26,427 Rooms (MOT)
Average Growth in Hotel Room Stock (2000-2024)	2.6% (MOT)
Total Number of Hotels (2024)	210 Properties (MOT)
Total Number of Workers in Accommodation & Restaurant Services (2024)	43,913 (MOT)
2025/2026 Ministry of Tourism Budget	95.5 Million USD (MOT)

Source: This table provides an overview of the scale of global, Caribbean, and Jamaican tourism. Data comes from the United Nations: World Tourism Organization (UNWTO),the Caribbean Hotel & Tourism Association (OECD), the Jamaican Ministry of Tourism (MOT), and the Organization for Economic Cooperation and Development (OECD). A room-night is a single overnight booking of a hotel room by an individual or party.

2 Context and Background

2.1 Jamaica and Tourism

Jamaica is an island nation located in the Northern Caribbean Sea, with a population of 2.8 million people. An upper-middle-income country, Jamaica's economy is heavily reliant upon tourism, which accounts for 10 percent of GDP directly, and for 30 percent of GDP when considering back-linkages (Mooney 2020). The tourism sector also accounts for about 30 percent of the labor force when measuring direct and indirect employment (Mooney 2020). Other Caribbean countries have comparable levels of specialization in tourism services, with tourism comprising an average of 25 percent of Caribbean GDP between 2015 and 2019, according to the OECD (OECD and Inter-American Development Bank 2024).

Following independence in 1962, the Jamaican government sought to speed economic development by its natural resources of "beaches and bauxite" (King 2001). The government created the Ministry of Tourism and instituted various laws such as the Tourist Board Act (Parliament of Jamaica 1969) and the Hotel Incentives Act (Parliament of Jamaica 1971) to support the development of the new industry.

Considerable investments were made in infrastructure and advertising, with the goal of attracting foreign direct investment. The geographic base of the industry is along Jamaica's northern and western coasts,[16] although in recent years there has been increasing development along the southwestern and northeastern coasts

[16]The unofficial capital of Jamaica's tourism industry is Montego Bay in the Northwest of the island. Montego Bay is Jamaica's second-largest city (behind the capital and industrial hub of Kingston) and is the point of entry for the vast majority of international tourists visiting the country for vacation (Jamaican Ministry of Tourism 2025).

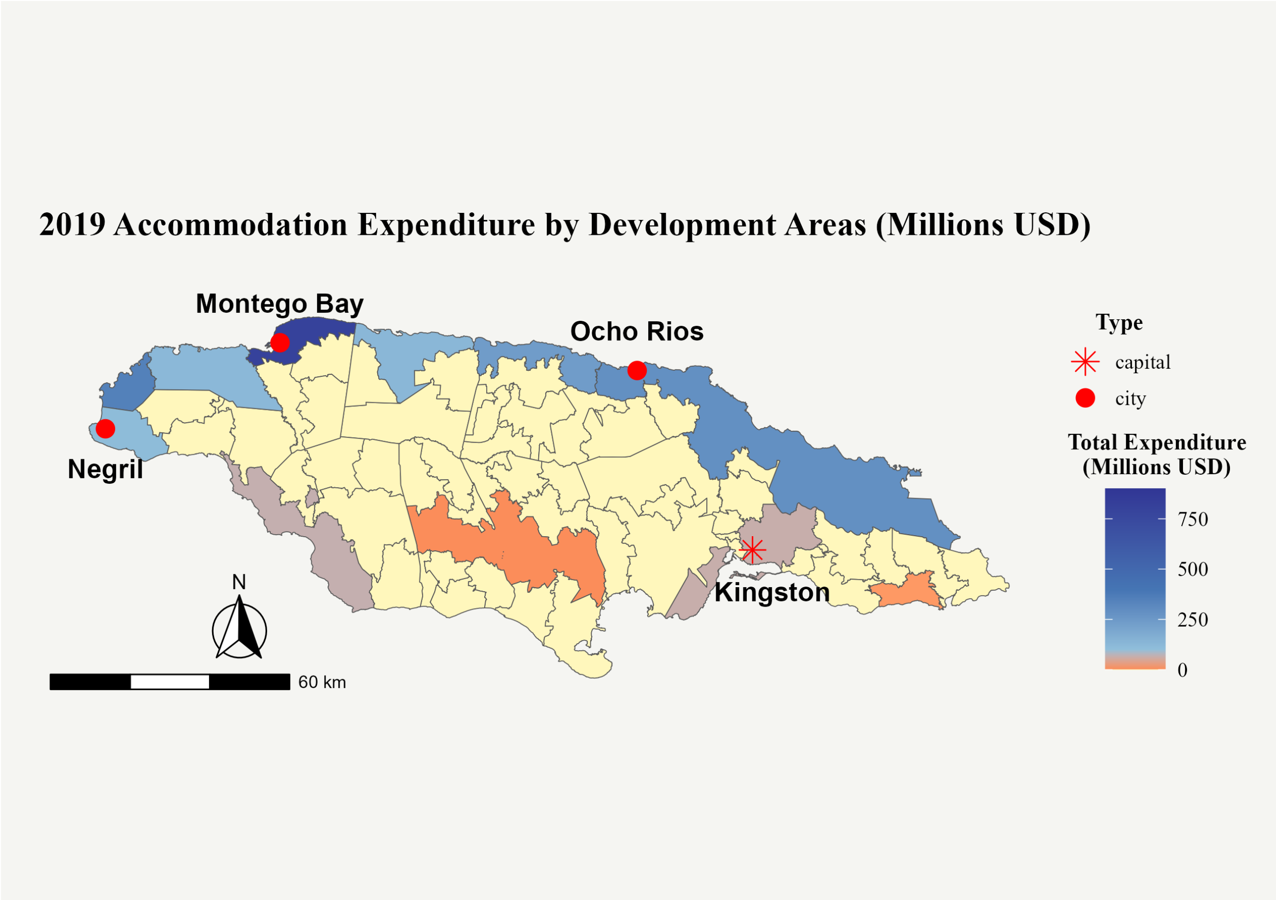


Figure 1: **Tourism Accommodation Expenditures Across The Island**

Figures estimated from the Jamaican Ministry of Tourism Exit Surveys, combined with annual arrival data, provide the MOT annual reports. All values are in 2024 U.S. Dollars. Yellow shading means that a particular development area received no revenue from accommodations in 2019.

as well. In 2019, over 75 percent of visitors stayed at accommodations located in Saint James, Trelawny, Saint Ann, Saint Mary, Westmoreland, or Hanover.[17]

Of the more than 4.2 million visitor arrivals in 2024, roughly 2.7 million were "stopover" visitors: those who spent at least 24 hours in the country (Jamaican Ministry of Tourism 2025). The remaining 1.6 million were visitors arriving on cruises, who typically spend less than a day in the country. Although stopover visitors made up about 56% of total arrivals to Jamaica, their spending made up more than 95 percent of total visitor expenditure, as can be seen in the MOT Annual Statistics publication for 2024 (Jamaican Ministry of Tourism 2025). Therefore, in order to understand the impact of tourism on economic development, it is reasonable to devote attention to the population of stopover arrivals.

As expected, the tourism industry is an important employer. According to the IADB, the tourism sector employed approximately $250,000$ Jamaicans in 2019 either directly or indirectly (Mooney 2020). This was

[17] *Source*: Jamaican Ministry of Tourism (2020). See figure 18 in the appendix for a map of Jamaica with the names of the parishes.

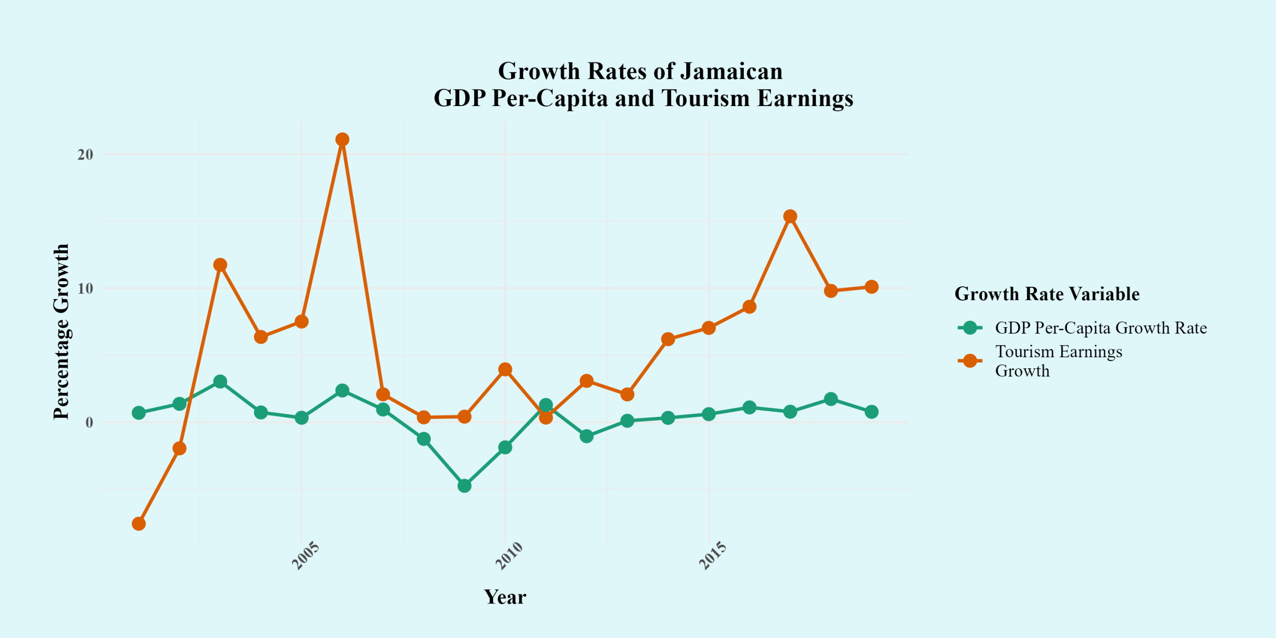


Figure 2: **Tourism vs. GDP Per-Capita Growth**

Source: World Bank Development Indicators. This graph shows a comparison of year over year growth rates for Jamaican tourism revenues and the growth rate of GDP Per-Capita.

roughly a quarter of the labor force at the time. Given spatial variation in the intensity of touristic activity and temporal variation in this intensity, there exists the opportunity to causally identify the impacts of tourism on Jamaican households.

The tourism sector has grown considerably over the last 60 years. Since 2000, arrivals have grown 5% per year on average and the room stock has grown at an average annual rate of 2.6%. However, the strong performance of the tourism industry has not translated into similar success in the larger economy. Over the same period, growth rates of real GDP per capita have oscillated between 1 and −1 percent. Value added per worker has been mostly stagnant, even as the tourism share of the labor force has grown and the industry has been marketed as the key to generating long-term national prosperity.

Jamaica's tourism product is based heavily around the "all-inclusive" resort model,[18] as is the case in many developing economies dependent on tourism (Tavares 2015). This resort model is characterized by a comprehensive package of accommodations, dining, and activities, restricted primarily to the resort premises (Issa and Jayawardena 2003). As a result, tourists often spend considerably less time and money in the communities in which resorts are located than when staying in other types of accommodations (Çiftçi et al. 2007). This is markedly different from destinations like Barcelona or Amsterdam, where visitors typically stay in hotels or short-term rentals located in areas popular with tourists (Allen et al. 2021), and spend money on attractions,

[18] According to the Ministry of Tourism in 2024, 87 percent of hotel room-nights sold were for all-inclusive resorts. All-inclusive resorts accounted for 79.5 percent of the hotel room stock (Jamaican Ministry of Tourism 2025).

Table 2: Household Characteristics: Urban - Rural Comparison

	Urban Households Mean	Urban Households SD	Rural Households Mean	Rural Households SD	Comparison T-Statistic
Per-Capita Consumption Expenditure	3602.86	3161.80	2682.49	2292.67	25.35***
Per-Capita Total Expenditure	3990.83	3983.195	2889.95	2762.260	24.26***
Per-Capita Food Expenditure	1613.62	1289.520	1355.71	1308.996	14.04***
Per-Capita Non-Food Expenditure	1990.36	2319.748	1327.30	1362.054	26.84***
Per Capita Non Consumption Expenditure	523.55	1487.020	292.83	1008.639	11.60***
Non-Food Share of Consumption Expenditure	0.52	0.145	0.48	0.135	24.67***
Non-Food Share of Tot. Expenditure	0.48	0.139	0.46	0.131	19.51***
Consumption Share of Tot. Expenditure	0.95	0.095	0.96	0.080	-12.44***
Observations	11196		18370		29567

Notes: All statistics are weighted by household size. All values are provided in 2024 U.S. Dollars. Non-Consumption expenditure relates to spending on services such as insurance, legal fees, and education. Total expenditure is equal to the sum of consumption expenditure and non-consumption expenditure. *** p<0.01, ** p<0.05, * p<0.1.

goods, and food directly in the local community.[19]

These differences in spending patterns and the structure of tourism may well result in very different welfare outcomes even from shocks of comparable magnitudes, which further supports the relevance of my study.

3 Jamaican Economy Background

3.1 Household and Labor Force Heterogeneity

Despite Jamaica's relatively small size, there is great variation in consumption levels across different regions of the country. Average real consumption expenditure per capita was 3021 USD for households during the study period; real urban consumption expenditures are on average 25 percent higher than rural consumption expenditures, as shown in table 2. There are statistically significant differences in the non-food share of expenditures, which is 4 percentage points higher in urban areas relative to rural areas.

4 Evaluating the Impact of Tourism

Rapid growth has been a key feature of the global tourism industry over the last two decades. The United Nations World Tourism Organization (UNWTO) estimates that the number of global travelers has increased from roughly 700 million 25 years ago to 1.3 billion in 2024 (World Tourism Organization 2024). According to the World Travel and Tourism Council, tourism made up roughly 10% of global GDP in 2019, accounting for 25% of all new jobs created worldwide (World Tourism Organization 2023). Following the collapse of the

[19]In 2024, visitors to Jamaica on average spent 61.1 percent of their total vacation expenditure on accommodations and food provided by their all-inclusive packages, 2.91 percent on outside food and beverage, 3.4 percent on shopping, and 12.7 percent on entertainment (Jamaican Ministry of Tourism 2025). On the other hand, in Barcelona in 2023, accommodations accounted for 47 percent of daily expenses, while food, attractions, and shopping accounted for 45 percent. *Source*:https://observatorituris me.barcelona/xifres-clau/.

industry during the COVID-19 pandemic, the global tourism sector rebounded to 80% of pre-pandemic levels in the first quarter of 2023, with roughly 235 million people traveli ng internationally — more than double the number during the same period in 2022 (World Tourism Organization 2023).

Tourism has a number of qualities that make it an attractive sector for specialization in a developing country. It is an excellent source of foreign exchange. As a labor-intensive sector, it is capable of absorbing a large number of unskilled workers (Nayyar et al. 2021). However, tourism is also characterized by a lack of linkages to other sectors and low scope for productivity growth (Nayyar et al. 2021), which are critical characteristics of industries that produce economic growth.

The Jamaican government's goals for developing the tourism sector are outlined in the "Vision2030" national development plan. The first three goals are that Jamaican tourism is "inclusive and facilitates broad participation by Jamaica," that the industry has "An adequate workforce within the sector that is skilled, educated and motivated," and finally that tourism is "A highly integrated sector which can act as a driver for economic development."

As noted above, roughly 83 percent of the Jamaican labor force is employed in low- and mid-skilled occupations.[20] Similarly, low- and mid-skilled occupations make up 82.3 percent of tourism-related occupations.[21], [22] Given that these occupations do not require more than a secondary level of schooling (International Labour Office 2012), and account for the largest share of occupations across Jamaica's sectors, it is reasonable to expect that Jamaican workers are largely mobile between tourism industries and other segments of the economy. This mobility implies that a positive shock to tourism may result in significant spillovers for a wide swath of the labor force in a particular development area.

An increase in the demand for local tourism services in an area is a positive shock to that area. The theoretical framework of Moretti (2010) interprets this as a sudden increase in labor demanded by tourist services, and predicts that this will produce an increase in employment in the local tourism sector. The framework also predicts that there will be increases in earnings for workers in the industry experiencing the demand shock, assuming that labor is mobile between sectors, and that there is imperfect labor mobility between regions. It predicts that the extent to which workers in other sectors benefit depends on the same factors and on whether or not the sector is tradable. Specifically, it predicts that the local non-traded sector will benefit, while the effects are unclear for the tradable sector.

What can be said about the geographic mobility of Jamaican workers? As in many developing countries, there has been considerable movement within Jamaica over the previous two decades from rural areas in the country's interior and south to urban centers. This has been driven by the familiar factors of rural poverty and the expectation of better job opportunities in urban centers. A large portion of this migration has been attributed to the development of tourism. According to the 2022 census, the fastest-growing parishes during

[20] Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions.

[21] The Statistical Institute of Jamaica defines tourism-related activities as sectors that heavily participate in the tourism industry, such as Hotels & Accommodations, Transportation and Logistics, etc.

[22] Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions Surveys.

the period 2011-2022 were St. James, Westmoreland, Hanover, and Trelawny, which are also major hubs of Jamaica's tourism industry. While this migration has been considerable, it has been on average much slower than the growth of Jamaica's tourism industry. This suggests that, while there is some mobility for Jamaican workers, there still remain significant frictions. One potential friction may be concerns about crime, as well as difficulties with procuring suitable housing as urban areas struggle to update their infrastructure.

What would be the effects of an influx of new residents into a development area in response to a labor demand shock? The new arrivals would generate an increase in demand for local services, especially housing. This might cause an increase in the prices of these services. In addition, an increase in the size of the local labor force would place downward pressure on wages. Whether these shocks also increase *real* consumption depends on the relative magnitudes of the changes in nominal earnings and changes in the cost of local goods and services (Moretti 2010; Aragón and Rud 2013; Allen et al. 2021; Almagro and Domínguez-Iino 2025). In the case of Barcelona, Allen et al. (2021) show that, while on average tourism lowers the welfare of Barcelona's residents because of increases in the cost of living, there is significant heterogeneity as to which residents have worse outcomes. In the Jamaican setting, where tourists are more segmented from the local population than in the urban tourism in Barcelona, I do not expect a significant effect of tourism on local food prices. I also would not expect a significant short- or medium-run effect on rent. Unlike settings where a large share of tourists stay in short-term rentals in close proximity to residents, tourists in Jamaica and other developing countries overwhelmingly stay in resorts, hotels, and resort-villas, as shown in 40 in appendix A.9.

The occupations and skill levels that benefit from tourism are predicted to depend on the skill makeups of the industries most impacted by the demand shock. Because low- and mid-skilled occupations make up the majority of Jamaica's tourism workforce and workforce overall, it would seem those groups would benefit most. Furthermore, because those in the lowest quartile of the consumption distribution are disproportionately likely to work in the mid- and lower-skilled occupations that dominate tourism, it is reasonable to expect that households in the lowest quartile would benefit most from increased tourism.

5 Data

At the heart of this analysis are two uniquely rich and informative datasets: Jamaican Ministry of Tourism (MOT) exit surveys of departing visitors, and nationally representative household expenditure surveys. The MOT administers these surveys each month to a random sample of tourists leaving through either of Jamaica's two main international airports. The exit survey provides a very detailed picture of tourist demographics, spending across more than 20 sub-categories, and, critically, the names of the hotels or hotel-like accommodations[23] in which a party stayed.[24] From these datasets spanning more than two decades, we can observe a remarkably thorough cross-section of the choices of stopover arrivals in Jamaica. The surveys also allow me to link accom-

[23]Hotel-like accommodations include resort villas, guesthouses, and resort-apartments.

[24]In the exit surveys, demographic questions include country and state/province of origin, the number of people in a travel party, their reason for visiting, length of stay, and income. Spending questions include the amount spent on hotel rooms, hotel food, groceries, transportation, restaurants outside the hotel, and attractions.

modation spending by tourists to specific Jamaican development areas based on the location of the hotels in which they stay. Utilizing publicly available aggregate arrival data, I am thus able to estimate what share of tourists choose to stay in particular areas of the country, the amount of spending on accommodations within these areas, and the relative contribution to local tourism revenues by tourists from different regions of origin in a given year. These exit surveys present a comprehensive picture of the spending behavior of stopover arrivals to Jamaica, which are the primary source of revenue for the Jamaican tourism sector. For the study period, I construct a repeated cross-section with roughly $80,000$ travel parties and $150,000$ individuals.[25]

I then pair this tourism dataset with an equally informative set of household data coming from the Jamaica Survey of Living Conditions (JSLC). The JSLC is an annual Living Standards and Measurement (LSMS)-style survey conducted on a representative sample of the Jamaican population. It is administered by the Statistical Institute of Jamaica (STATIN). The JSLC uses a two-stage stratified random sampling design. The modules cover a wide range of topics related to household well-being, such as expenditure across different types of consumption, education levels, health, and labor force participation. In most years, around 2000 households are surveyed, which results in around 6000 individuals being included in the sample, or about 0.3 percent of the Jamaican population. I construct a repeated cross-section of 30680 households, representing 98883 individuals. The data covers the years 2001-2004, 2006, 2008-2011, 2013-2014, 2016-2019 and 2021.

In order to support findings from the baseline repeated cross-section specification, I also construct an unbalanced panel from a subset of the Jamaica Survey of Living Conditions data, following the methodology used by Handa (2007) to analyze household movement across the income distribution. Prior to 2018, the JSLC would occasionally re-sample households from previous years, using the same sampling frame. On average, about half of the households in one year may have been re-sampled. This the case for the years 2002-2003, 2004-2007, 2008-2010, 2013-2014, and 2015-2016. Following the Handa (2007) methodology, I am able to construct an unbalanced panel of approximately 6000 Jamaican households representing some 30,000 individuals.[26]

I make use of administrative shapefiles for the 2001 and 2011 censuses to link touristic activity and households across space. The shapefiles provide the boundaries of the 86 development areas and 5776 enumeration districts[27] into which STATIN divides the country for data collection and analysis. Development area boundaries are designed to encompass municipalities with similar economic and social characteristics.[28] Therefore, development areas represent a useful level at which to estimate variation in tourism activity. Development areas include urban centers as well as rural hinterlands, as can be seen in figure 3, showing the Greater Montego Bay development area and its urban and rural sub-districts.

[25] I scale the estimates of local tourist accommodation expenditures by publicly available aggregate tourism statistics published by the Ministry of Tourism, based on the customs forms filled out by arriving tourists.

[26] Analysis by Handa (2007) showed that the full sample of households in the JSLC and the panel sub-sample do not exhibit statistically significant differences in poverty rates, providing further confidence that these two datasets are statistically equivalent.

[27] Enumeration districts are the smallest geographic units used for data collection during the census by STATIN. I use "enumeration district" interchangeably with "district(s)" and "sub-district(s)."

[28] The Social Development Commission describes a Development Area (DA) as "a grouping of communities based on geographic, demographic, economic, and social criteria/commonalities. The DA has the potential for growth to satisfy the needs of the people. The DA has a centre or hub to which people gravitate for socio-economic activities. The DA is usually given the name of the community which is the hub of activities for that area."

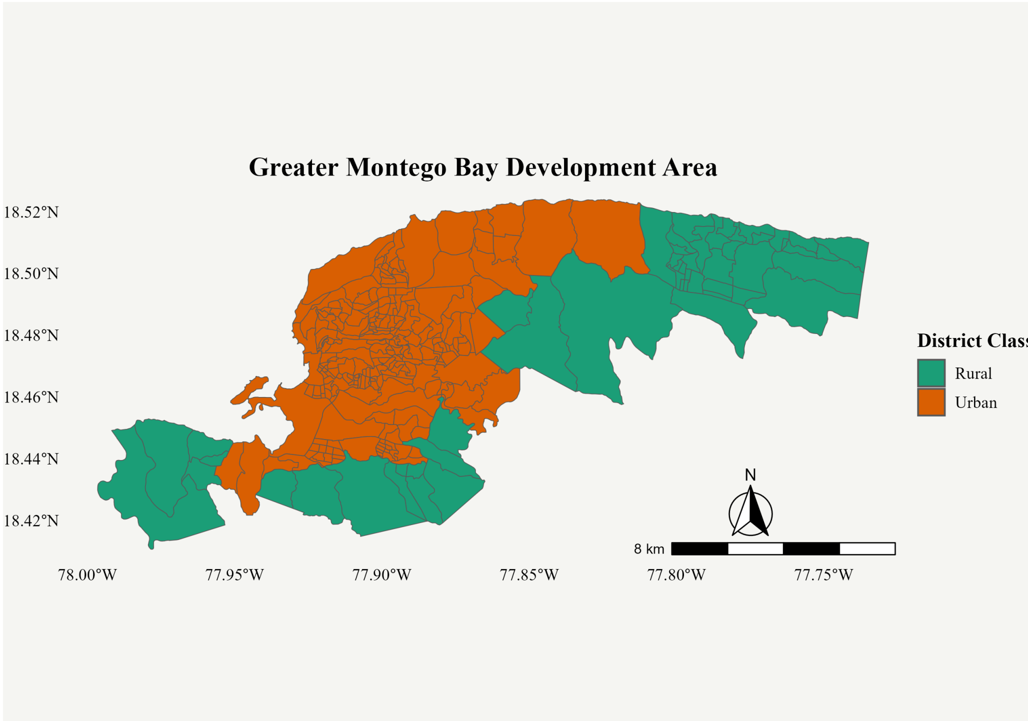


Figure 3: **Greater Montego Bay Development Area**

Notes: Urban enumeration districts are shaded in orange, while rural enumeration districts are shaded in green.

Sampling for the census occurs at the level of the enumeration district. The district boundaries for each census are maintained for the following decade until the next census. While I do not observe development areas for the 2001 census, I am able to achieve spatial consistency by taking advantage of the enumeration district boundaries. For years prior to 2011, I overlay 2011 development area boundaries on 2001 enumeration districts in order to obtain consistent spatial units.[29] I aggregate some neighboring development areas into one unit if one or both have only low to moderate tourism levels.

6 Empirical Approach

The structural equation describing the relationship that we wish to estimate is given below. In this equation, y_{it} represents per-capita expenditures for household i in development area d in year t. The term $Tourism_{dt}$ represents total tourist accommodation expenditures[30] in development area d for the same period. X_{idt} is a vector of household characteristics, α is a constant, and ϵ_{idt} is an idiosyncratic error term. The equation is the following:

[29] If an enumeration district is not wholly contained within a particular development area, I assign the households in that district to the development area where the largest share of the district is located.

[30] I choose to estimate tourism intensity with only accommodations expenditures because I can guarantee that this expenditure was spent on services received within a given development area. On the other hand, expenditures on restaurants outside the hotels, clothing, attractions, etc. may have occurred in other localities.

$$Y_{idt} = \alpha + \beta Tourism_{dt} + \psi X_{idt} + \gamma_t + \lambda_d + \epsilon_{idt}. \tag{1}$$

The earlier predictions from the literature serve as useful guides for empirically analyzing the effects of tourism on household welfare for Jamaican communities. The characteristics of the Jamaican tourism sector and labor force, and insights from studies analyzing the effects of local economic shocks (Moretti 2010; Allen et al. 2021; Aragón and Rud 2013), inform four testable hypotheses for this analysis.

- H1: Increases in development area tourist expenditures will result in an increase in real per-capita consumption for households in that development area.
- H2: The increase in real per-capita consumption will be greater for households employed in tourism-related industries than for those employed in other sectors.
- H3: Households in low- and mid-skilled occupations will experience the largest increase in real consumption.
- H4: The benefits of increases in tourism activity will accrue primarily to those in the lowest quartile of the per-capita expenditure distribution.

Because development areas are collections of economically linked communities, hypothesis one predicts that an influx of tourists to a particular development area d will result in higher real consumption levels for d's households relative to households in areas that do not experience an increase in tourism revenues. The prediction of hypothesis two stems from the fact that an increase in tourism revenue for a development area is expected to increase the tourism industry's demand for labor and thereby increase wages within the sector (Moretti 2010). We would also expect there to be a smaller increase in real consumption for households employed in non-tourism industries, such as non-tourism services and manufacturing. The size of this increase is expected to depend on the increased demand-side competition in the labor market resulting in higher wages, the level of the increase in consumption of local goods and services by households owing to greater earnings, and the strength of back-linkages between tourism and other sectors. The prediction of the third hypothesis — that low- to mid-skilled workers will benefit most from tourism — is informed by the fact that households in these occupation classes make up the largest share of tourism workers. In addition, low- and mid-skilled workers are more likely than high-skilled workers to fall within the first consumption quartile; therefore, hypothesis four predicts that we will see the greatest increase in real consumption among households in that segment of the consumption distribution.

6.1 The Shift-Share Instrumental Variable Approach

The level of tourism in a given development area is not random. It is therefore likely that $Tourism_{dt}$ is correlated with our error term. For example, areas that have higher levels of tourism revenue may also have qualities that attract more affluent households to live there, or that lead to better economic outcomes for locals. The endogeneity of $Tourism_{dt}$ motivates the use of a shift-share instrumental variable (SSIV), which will provide variation that is orthogonal to the attributes of the development area and that may influence household welfare.

This study joins an extensive body of research that employs SSIV identification strategies to correct for endogeneity. Since the approach was first introduced in Bartik (1991), in a study of the growth rate of employment in specific sectors, SSIVs have proven extremely versatile in a variety of areas, including trade and labor (Autor et al. 2013; Hummels et al. 2014), health (Miguel and Kremer 2004), migration and labor market outcomes (Card 2009), and the welfare impacts of location classification (Diamond 2016).

In constructing and evaluating the SSIV, I adapt the strategy of Allen et al. (2021), exploiting variation in which tourists from certain regions of origin choose to stay in Jamaica and how much they spend. The identification strategy of this paper follows the exogenous shock-based approach of Borusyak et al. (2022), as opposed to the exogenous share-based approach (Goldsmith-Pinkham et al. 2020). Several papers in recent years have suggested detailed guidelines to ensure accurate shift-share inference in the presence of exogenous shocks; a panel of exogenous shocks is relevant to this case (Borusyak et al. 2022; Borusyak et al. 2024; Borusyak et al. 2024; Adão et al. 2019). Regarding accurate calculation of standard errors in SSIV designs, Borusyak et al. (2024) and Adão et al. (2019) have developed techniques to calculate standard errors that account for the correlation in the residuals of units exposed to similar shocks. Adão et al. (2019) demonstrate that failure to account for residual correlations can lead to over-rejection of the null hypothesis. I utilize their approach in order to adjust the standard errors for correlation in the shocks and include these regressions in appendix A.1.

In the shift-share equation, $g_1, ..., g_k$ represent shocks common to all units, while $s_{i1}, ...sik$ are the exposure shares that vary across units. A shift-share instrumental variable takes the form:

$$z_i = \sum_{k=1}^{K} \underbrace{s_{ik}}_{\text{Share}} \underbrace{g_k}_{\text{Shift}}, \tag{2}$$

with the final instrument z_i being a share-weighted average of the shifts.

This shift-share instrumental variable identification strategy is best motivated through the logic of a shift-level idealized experiment. We can consider a hypothetical experiment in which we address the unobserved variables in development areas d that are likely to bias an ordinary least squares estimation of 1. Imagine that we implement a lottery that in each year randomly selects a certain number T of tourists from specific sending regions r to receive free flights to Jamaica. We are able to dictate that only people who are invited to Jamaica via the lottery are able to visit; thus, the aggregate number of tourists visiting Jamaica, T, and the number from each sending region, T_r, are determined entirely via the lottery. We can expect that regions from which a larger number of people are selected are likely to have more visitors to Jamaica relative to regions for which there are fewer awardees. Visitors then decide where on the island to stay, as well as the amount they spend on accommodations and other goods or services. Thus, only the number of tourists from each sending region is randomly determined. Because of this random assignment of awards, the shifts from year to year in the total spending on accommodations by tourists from specific sending regions, g_{rt}, are unrelated to development area d attributes, such as beach quality, which may influence our household outcome variables of interest y_{idt}.

Our goal is to approximate the experimental ideal explained above with existing variation in the data. I exploit two facts about tourism in Jamaica to proxy for ideal shifts and achieve identification. The first is that tourists from different regions of origin vary both cross-sectionally and over time in the areas of Jamaica they prefer to visit. The second is that tourists from different sending regions visit Jamaica in different magnitudes from year to year. The exposure share s_{drt} of each development area d to tourists from a sending region r in year t is the fraction of accommodation revenues in that development area earned from sending region r visitors. The shifts will be the changes in total accommodation spending by tourists from that sending region r in Jamaica overall. Jamaica's main tourism markets are the United States, Canada, and the United Kingdom, with secondary markets including the Caribbean, continental Europe, and Latin America. For the United States and Canada, the exit surveys also provide information on the states or provinces from which tourists are visiting. I divide tourist arrivals across seven regions given by the vector $r \in$ (Northeastern U.S., Western U.S., Midwestern U.S., Southern U.S., Canada, U.K. & Europe, Other Countries).

How do we justify the argument that year-to-year variation in spending by tourists from specific sending regions is uncorrelated with the error term? First, we can note that there was substantial year-to-year variation over the period 2000 to 2021 in aggregate arrivals from these regions, as well as variation in where members of these groups tended to visit in Jamaica. This variation is driven by a combination of changing local economic conditions in the origin regions, changes in access to Jamaica via available air routes, shifting preferences, and possibly other factors. Some of this variation can be seen in figure 4 above, and in figure 17 in appendix A.9 .

Second, there was significant volatility in the global economy during the study period as a result of geopolitical and economic events, such as the September 11th terrorist attacks, the 2008 global financial crisis, and the COVID-19 pandemic. Decisions about whether and where to travel are influenced by factors such as inflation and debt crises (Khalid et al. 2020), risk preferences and demographics (Karl 2018), and the distance to a destination (Sirakaya and Woodside 2005). These shocks produced changes in travel and vacation patterns, and these changes may have differed depending on economic and political characteristics of particular regions of origin.

Thus, we obtain year-to-year variation in the arrivals of tourists from different regions of the world. Coupled with the differential levels of exposure of Jamaican development areas, these changes in sending region arrivals generate variation in local accommodations expenditures that is orthogonal to the characteristics of those areas that may influence the outcome variables of interest.

The instrumental variable below, based on tourist region of origin, $(g1, ...gk)$, utilizes shifts that are common to all units (development areas). The vector $(s_{i1}, ..., s_{iK})$ comprises the exposure shares that vary across units. The term $Tourism_{dt}$ represents total annual tourist accommodation spending in development area d, in year t.

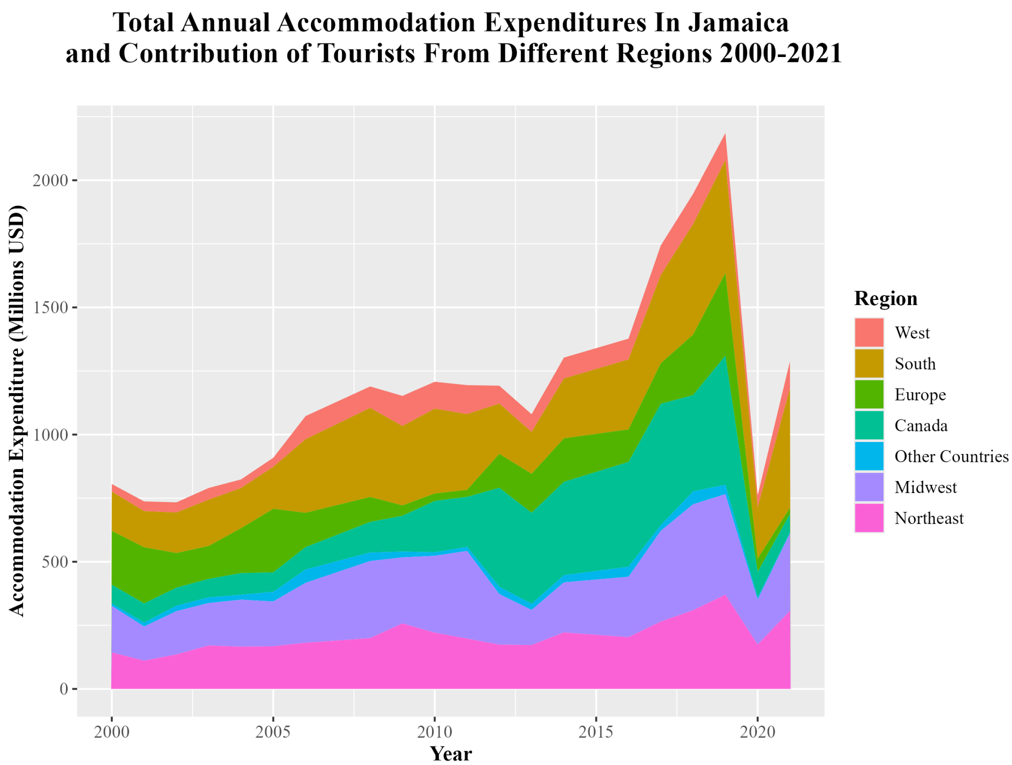


Figure 4: **Level of Expenditures By Sending Region**

*Notes:*Data comes from the Ministry of Tourism exit surveys. The contribution of tourists from various sending regions to total accommodations expenditures in Jamaica from 2000 to 2021 are shown. Expenditures are provided in 2024 Millions of U. Dollars.

This is equal to the sum of each region of origin r's expenditure on accommodations in the development area in year $t : Tourism_{drt}$.

$$Tourism_{dt} = \sum_{r \in R} \Big[Tourism_{drt} \Big] \tag{3}$$

Area d's exposure to region r tourists in year t is given by

$$s_{drt} = \frac{Tourism_{drt}}{\sum_r Tourism_{drt}} = \frac{Tourism_{drt}}{Tourism_{dt}} \tag{4}$$

With all development areas being a part of the set $\mathbf{D}$, total spending in Jamaica on accommodations by tourists from region r in year t is therefore:

$$T_{rt} = \sum_{d \in \mathbf{D}} \Big[Tourism_{drt} \Big]. \tag{5}$$

We can then define the "shift" — the change in total expenditures by tourists from region r between period $t = 1$ and period $t = 0$ — as:

$$g_{r1} = \frac{Tourism_{r1} - Tourism_{r0}}{Tourism_{r0}}. \tag{6}$$

The shift-share instrument therefore is given by:

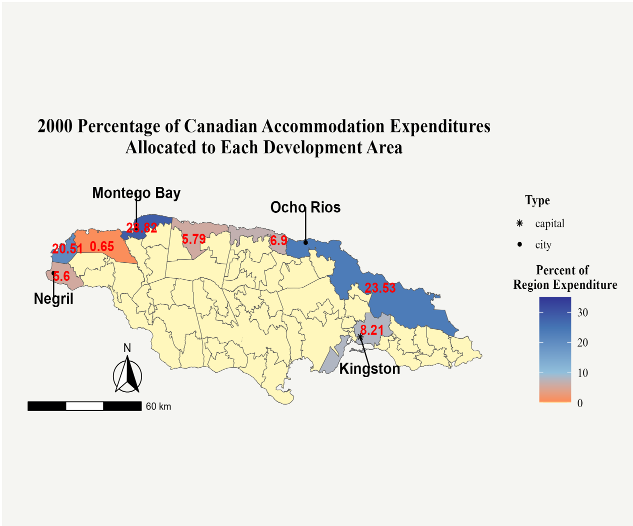
$$z_{dt} = \sum_{r \in R} s_{drt} \left[\frac{Tourism_{r1} - Tourism_{r0}}{Tourism_{r0}} \right] = \sum_{r \in R} s_{drt} g_{r1}. \tag{7}$$

The shifts are constructed using "leave-one-out" construction. In this approach, discussed and employed throughout the shift-share literature (Borusyak et al. 2022; Goldsmith-Pinkham et al. 2020; Autor et al. 2013), the shift-share instrument is written:

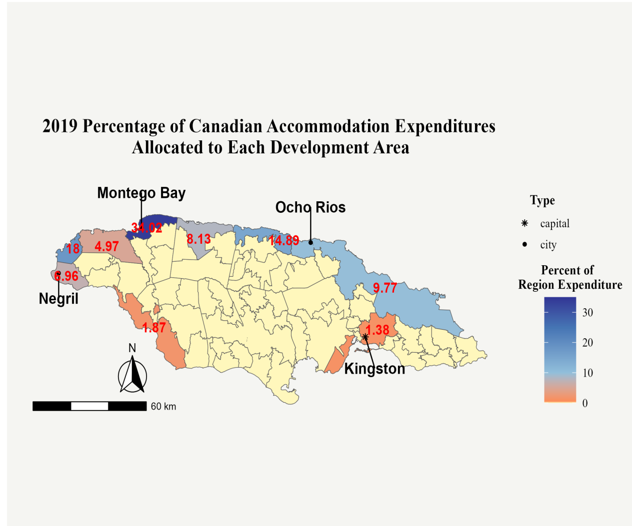
$$z_{dt} = \sum_{r \in R} = \sum_{r \in R} s_{drt} g_{r1,-d}, \tag{8}$$

meaning the calculated change in total spending in Jamaica by region r tourists does not include the group's change in spending for area d. This construction is meant to avoid bias in the instrument for area d, potentially caused by including area d shifts. Autor and Duggan (2003) show that including own-unit shifts could lead to a substantially stronger instrumental variable, although Goldsmith-Pinkham et al. (2020) show that the leave-out correction has a relatively minor effect when shifts are averaged over many observations.

Over time, tourists from different regions of origin have shifted the frequency with which they visit particular



(a) Canadians 2000



(b) Canadians 2019

Figure 5: Variation Over 2 Decades In Spatial Distribution of Canadian Tourist Expenditures

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from Canada between the years 2000 and 2019. The numbers show the share of total Canadian accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Canadian tourists in that particular development area in a given year.

areas of Jamaica. For example, in figures 5a and 5b, we can see variation in which parts of the island Canadians preferred to visit between 2000 and 2019. I show the same figure for tourists from the Southeastern United States in the appendix.

For the 2SLS specification, Y_{idt} will be either per-capita expenditure, or an indicator for poverty status (0: Above Poverty Line, 1: Below Poverty Line) for household i, in development area d, in year t, measured in both logs and levels. Tourism intensity is given by $Tourism_{dt}$, and is calculated as total expenditures on accommodations in a development area in a given year, while Z_{dt} is the shift-share instrument, and η_{idt} is an idiosyncratic error term for the first stage. Second-stage terms are the same as stated in the structural equation.

Stage 1:

$$Tourism_{dt} = \chi + \phi Bartik_{dt} + \iota X_{idt} + \omega D_t + \pi C_d + \eta_{idt} \tag{9}$$

Stage 2:

$$Y_{idt} = \alpha + \beta Tourism_{dt} + \psi X_{idt} + \rho D_t + \lambda C_d + \epsilon_{idt}, \tag{10}$$

Within the vector of household controls, X_{idt} is the number of members in a household, the sex of the household head, and whether or not the household is located in a rural enumeration district. Because the identifying variation occurs at the development area level, and because I expect the residuals of households within the same development areas to be correlated, I cluster the standard errors at this level.

6.2 Identification Checks

The key identifying assumption in using this shift-share instrument is that year-to-year variation in the accommodations expenditures of tourists from specific regions of origin r visiting Jamaica are uncorrelated with unobserved characteristics of the development areas or the households within them that may impact our outcome variables of interest. That is, whatever factors are driving the changes in spending by different tourist groups, and/or their decisions on where to spend their vacations in Jamaica, are not correlated with the unobserved characteristics of households and development areas. Formally, this can be represented as

$$\mathbb{E}[Z_{dt}\epsilon_{idt}] = 0 \tag{11}$$

In order for the instrument to be valid, it must satisfy relevance and the exclusion restriction. The first-stage regressions for the baseline findings show that the instrument functions well, with a first-stage of 29 for the specification including all controls and dummy variables. This strength is also reflected in graphs of the correlations shown in the appendix.

In order to satisfy the exclusion restriction, the instrument must affect household welfare only through tourist expenditures on accommodations, conditional on controls. This would be violated if the instrumental variable has an impact on per-capita expenditures through channels such as cost of living. For example, the instrument may also be correlated with higher per-capita expenditures through its effect on the prices of locally produced services as a result of tourist spending in the community. To account for such a channel, I inflate or deflate expenditures according to Jamaican regional price indices. Because I also normalize all expenditures to 2024 US dollars, the expenditure outcome variables capture real per-capita spending behavior and should not reflect price changes induced by local tourism activity.

I also use an Instrumental Variable Quantile Regressions (IVQR) to more precisely estimate the effects of shocks to development areas for households across the expenditure distribution. Given that the main dataset is a repeated cross-section, the decile within which a household falls is likely endogenous; IVQR provides a more accurate representation of the distributional impacts of the tourism shocks. I utilize the method based on Kaplan and Sun (2017), and perform the regressions with the associated SIVQR command in Stata (Kaplan 2023). Because this method utilizes a distance minimization technique that requires a sufficient number of observations for each quantile conditional on controls and dummy variables, I am unable to estimate the quantile regression with the full set of right-hand side variables from the 2SLS specification. Instead, I employ dummy variables for each of Jamaica's 14 parishes, and a dummy variable indicating whether the observation is before or after the year 2010.

6.3 Interpreting Beta

Let us now consider the possible implications of migration in our interpretation of β. We may be concerned that β captures the effects of both existing local populations and Jamaicans who migrate to an area in response

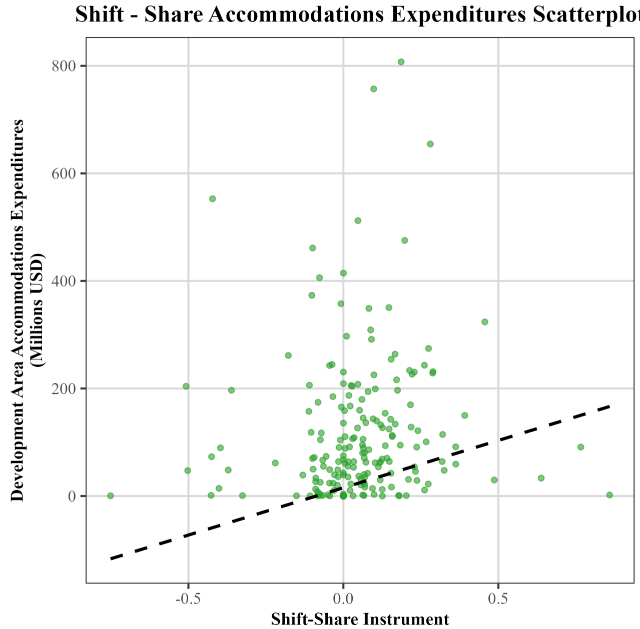


Figure 6: **Scatterplot of the Shift-Share IV and Tourist Accommodation Spending**
This scatterplot shows the relationship between the shift-share instrumental variable and the endogenous regressor: Development Area-level accommodations revenues.

to the labor demand shock. We may also be concerned that the effects of tourism earnings are not captured in cases where a worker migrates from a rural area or a community with no tourism and then uses earnings from tourism to provide for family in that non-tourism producing area. This would imply that the $Tourism_{ct}$ variable for a non-tourism producing area c is understated. Therefore, $\hat{\beta}$ would be attenuated by this under-counting of development area tourism revenues. We can thus interpret the estimates of β as a lower bound of the true effect of tourism revenues on households.

6.4 Shift-Share Instrument Diagnostics

One concern when employing a shift-share instrumental variable identification strategy is that we do not appropriately account for correlation between the shocks (Borusyak et al. 2024; Adão et al. 2019). Valid inference must account for the possibility that similar units, d, with similar shares s_{dr}, may have correlated ϵ_d. In this setting, such correlation may arise if development areas that cater to tourists from the same sending regions are exposed to the same unobserved sending region shocks. This could lead to over-rejection of a true null hypothesis, as shown by Adão et al. (2019). Adão et al. (2019) introduce a variance estimator that corrects for this potential over-rejection problem.[31] When I correct my baseline estimates with their approach, there is no significant change in the findings. I show a comparison of my main estimates with different standard error approaches in table 13 in appendix A.2.

Another concern for shift-share inference when working with panels of units is the potential for correlation

[31] The estimator of Adão et al. (2019) is asymptotically valid no matter the correlation of errors across observations, under the condition that shocks are mutually uncorrelated — that is, $g_r \not\sim g_s$.

in the shocks across periods. If such correlation is present, then past shifts may influence both present shares and the outcome variable. If this is not accounted for, then the estimation will suffer from omitted variable bias and cannot be viewed as a valid natural experiment. One way to correct for this is to extract the idiosyncratic component of each shock before constructing the instrument Borusyak et al. (2024). I regress each sending region shock on the interaction of region and development area controls, and then use the residuals as the new shocks in calculating the shift-share instrument. Graphs comparing the two versions of the shocks are available in appendix A.1. The final shift-share instrument and the endogenous regressor are shown in figure 6.

7 Results

I now present the main findings and their relationship to the four hypotheses listed above. I first describe the baseline findings, along with results for the analysis of outcomes related to household well-being. The following section will present results related to the effects of tourism on various segments of the labor force. I will then present results of the quantile regression describing the impacts of tourism across the household expenditure distribution. Finally, I will analyze the results of regressions evaluating the plausibility of alternative explanations.

7.1 Baseline Findings and Welfare Outcomes

Table 3 displays findings for regressions addressing the foundational question of whether increases in tourism revenues produce real increases in per-capita consumption for local households. Log per-capita expenditure is the outcome variable for these regressions. The regression results are separated by whether households reside in an urban sub-district or a rural sub-district. The F-Statistic for the urban sample regression is the strongest at 70, followed by the full sample F-Statistic at 53, and the rural F-Statistic of 22. The coefficient on tourism expenditures for the urban sample is 0.027, and is statistically significant at the 1 percent level.

Hypothesis 1 is that increases in development area tourist expenditures will result in an increase in real per-capita consumption for households in that development area. The log-level specification of the regression implies that an increase of 10 million USD in annual real tourism revenues in a development area increases real per-capita expenditures by 2.7 percent for households in the development area, relative to households in areas that do not receive this shock. There is no statistically significant effect of increased tourism earnings on rural households, as shown in column 2. Though the first stage of the rural regression is relatively weak, the full-sample regression in column 3 lends support to the results from column 2, given the stronger full-sample first stage. The coefficient on tourism expenditures is again insignificant for rural areas in the regression utilizing the entire sample.

Is the effect for urban households economically significant within the context of Jamaica? This depends in part on the typical magnitude of the variation in development area tourism expenditures. The median change in tourism levels for a development area from one year to the next is 7.92 million USD, and over the course

Table 3: IV: Relationship Between Tourism Earnings and Log Per-Capita Household Expenditure

	Annual Log Per-Capita Expenditure		
	Urban Areas (1)	Rural Areas (2)	Rural and Urban (3)
Panel A: Second Stage			
Tourism Expenditure (Tens of Millions USD)	0.027***	-0.003	0.011
	(0.005)	(0.010)	(0.007)
	[.01429, .03908]	[-0.024,0.017]	[-0.003,0.026]
Observations	11447	19163	30610
Number of Clusters	34	57	59
Bootstrapped Hypothesis Test	Yes	No	No
Panel B: First Stage			
Shift-Share Instrument	11.889***	11.882***	11.920***
	(1.418)	(2.544)	(1.633)
	[9.109,14.668]	[6.896,16.868]	[8.719,15.122]
First-Stage F-Statistic	70	22	53
Observations	11447	19163	30610

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 4: IV- Effect of Tourism On Selected Measures of Household Well-Being -Urban Households

	Log-Expenditure Outcomes			Level Welfare Outcomes		
	Log Per-Capita Food (1)	Log Per-Capita Non-Food (2)	Log Per-Capita Non-Consumption (3)	Per-Capita Healthcare (4)	Per-Capita Education (5)	Loan Repayment (6)
Tourism Expenditure	0.023***	0.029**	0.023	19.770**	-1.443	24.393
(Tens of Millions USD)	(0.003)	(0.010)	(0.033)	(5.656)	(4.498)	(9.746)
	[.01853, .02983]	[.003635, .05586]	[.08037, .0884]	[4.143, 35.02]	[11.94, 11.29]	[7.233, 43]
First-Stage F-Statistic	70	72	77	70	70	71
Observations	11438	11432	7217	10599	10599	6951

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95 percent confidence interval corrected with a wild bootstrap. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

of five years is 31.06 million USD.[32] Therefore, the median development area in Jamaica should see per-capita consumption levels increase by 2.1 percent on average in response to increases in tourism activity. Over the course of a five-year period, increases in local tourism intensity would produce an 8.3 percent increase in real per-capita expenditures on average for households in the area. The average urban household's annual per-capita consumption expenditure increases by 97 USD as a result of an increase of 10 million USD in tourism earnings for the development area. This can be compared against the average .2 percent growth of GDP per capita in Jamaica over the study period. Thus, the first hypothesis, that greater tourism earnings generate greater consumption, is shown to be true, but only for urban Jamaicans.

Considering only the changes in real per-capita expenditures may inaccurately capture variation in household welfare. In order to more holistically capture household well-being and living conditions, I replicate the baseline regressions with disaggregated expenditure categories as the left-hand-side variables. The first three regressions consider food, non-food, and non-consumption expenditures. The fourth, fifth, and sixth regressions deal with three important categories of household well-being: health, human-capital investments, and financial security. Together, these regressions provide a more complete picture of how consumption is changing, and can provide additional insight on the extent to which the findings can be interpreted as changes in household well-being.

Table 4's first and second columns show that, for urban households, the coefficient on tourism expenditures is positive and statistically significant for real food and non-food consumption. The coefficient in the food regression is significant at the .1 percent level; the coefficient for the non-food regression is significant at the 5 percent level. In terms of magnitude, individual food expenditures increase by 2.3 percent on average per 10 million USD in area tourism revenues, while individual non-food expenditures increase by 2.9 percent on average. The difference between these coefficients is not statistically significant, indicating that growth in local tourism revenues produces equivalent percentage increases in real food and non-food spending by locals.

[32]Source: Based on calculations from the Ministry of Tourism exit survey. See table 41 in appendix A.9 for selected summary statistics on year to year variations in tourism.

Table 5: IV: Relationship Between Household Poverty and Development Area Tourism Revenues

	Likelihood of Household Being Below the Poverty Line		
	Urban (1)	Rural (2)	Full Sample (3)
Tourism Expenditure (Tens of Millions USD)	-0.006**	0.002	-0.002
	(0.002)	(0.004)	(0.002)
	[-0.010, -0.002]	[-0.008, 0.011]	[-0.007, 0.003]
First-Stage F-Statistic	70	22	53
Observations	11448	19164	30612

Notes: The outcome is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45, as is the case for the urban sample. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Although the percentages are statistically equivalent, the value of the increases in real spending are not equivalent for each of these categories. Because the average per-capita annual expenditure by urban households is 1613 USD for food and 1900 USD for non-food goods, the dollar value of the increased expenditures is 37 USD for food and 57 USD for non-food goods. Approximately 61 percent of the increase in average consumption is spent by households on non-food items, indicating that exposed households use much of their additional consumption to spend beyond basic food items. Column 3 of table 4 shows that there is no change on average in per-capita non-consumption expenditures resulting from increased tourism activity.

The fourth column of table 4 uses per-capita expenditure on medical services and products as the outcome variable. An annual increase of 10 million USD in tourism receipts for an area increases real expenditure on health goods and services by 19 USD for Jamaicans in urban sub-districts. This is statistically significant at the 5 percent level. Columns 5 and 6 use per-capita expenditures on education goods and services and annual household expenditures on loan repayments and interest payments; the coefficients on tourism revenues for both are statistically insignificant. These results indicate the validity of the first testable hypotheses for urban contexts, and in particular suggest that tourism growth is improving the welfare of households in the immediate vicinity of tourist zones. Increased revenues result in higher real consumption levels for both food and non-food items, with the majority of this increased consumption going to non-food goods, and approximately one-third of the increase in non-food expenditures being used by households to increase consumption of medical goods and services.

Table 5 shows that increased tourism earnings for a development area also reduce the likelihood of a household falling below the poverty line in that area; however, once again, this effect is only evident for urban dwellers. An increase in tourism earnings of 10 million USD results in a statistically significant .6 percent lower likelihood of an observed urban household being below the poverty line relative to other development areas. Taken together with the results on individual consumption categories, these results partially support the first testable hypothesis: urban households do derive welfare benefits from increases in their community's tourism levels, with real gains in consumption level, purchasing of health-related services, and a lower likelihood of poverty. It is not clear, however, that these same benefits extend to rural households within the same development area. For rural areas, there are no observed changes in any categories of real consumption, or in the likelihood of poverty. At the same time, the shift-share instrumental variable is relatively weaker when evaluating the rural sample. Because most rural households reside in areas with little to no tourism, the lack of an effect of increased tourism earnings for rural households can be interpreted with some caution.

7.2 Heterogeneous Impacts in the Labor Force

The next two hypotheses address questions of which workers benefit from tourism shocks. Hypothesis 2 is that the increase in real per-capita consumption is greater for households employed in tourism-related industries. Hypothesis 3 is that households in low- and mid-skilled occupations will experience the largest increase in real consumption.

In table 6, I present results based on whether or not a household is employed in a tourism-related industry. In order to confirm the predictions on employment in the literature, in the first column I regress an indicator variable that takes a value of 1 if a household works in the tourism industry on tourism earnings in the development area. As expected, the effects are positive and statistically significant. Columns 2 and 3 repeat the baseline regression, but column 3 includes only the sample working in non-tourism industries, and column 3 is only made up of the sample of tourism workers. In contrast to the predictions of the second hypothesis, only households in non-tourism sectors experience increases in real consumption levels as a result of the increase in tourism. The coefficient of 2.9 percent implies that the baseline findings are driven entirely by non-tourism workers. Similarly for poverty, columns 4 and 5 reveal that the effects of tourism on poverty only occur for non-tourism workers. Thus, while employment in tourism certainly increases, consumption levels do not appear to be increasing in households that work in tourism. The first stage of the tourism worker sample regression is not as strong as that of the non-tourism sector, in part because of fewer observations when limiting the sample to the tourism workforce alone. Thus, it may be best to interpret these findings as confirming an effect in non-tourism sectors but being unable to confirm an effect within the tourism industry, rather than as a definitive indication that tourism workers are not benefiting directly.

The third hypothesis predicts that, because of the large share of low- and mid-skilled workers in the tourism sector, these workers will benefit the most from a positive shock to tourism in a community. Table 6 repeats the baseline regressions for separate samples of different occupations across the different skill levels for urban households. The first column shows that, for unskilled occupations, there is no effect of tourism on annual real per-capita expenditures. It is occupations in the level 2 skill segment that are benefiting the most, namely, "Skilled Trades" and "Plant and Machine" operators. This is partly in keeping with the third hypothesis: mid-skilled workers benefit, but low-skilled workers do not. There is no statistically significant effect for skilled agricultural jobs, which is consistent with the lack of observable findings for rural households. The skill level 3

Table 6: IV - Results By Employment in Tourism-Related Industries - Urban Households

	Employment	Real Log Per-Capita Expenditure		Likelihood of Being The Below Poverty Line	
	Likelihood of Employment in Tourism	Non-Tourism Related Industry	Tourism Related Industry	Non-Tourism Related Industry	Tourism Related Industry
	(1)	(2)	(3)	(4)	(5)
Tourism Expenditure	0.010**	0.029***	-0.010	-0.006**	-0.006
(Tens of Millions USD)	(0.003)	(0.005)	(0.010)	(0.002)	(0.004)
	[0.003,0.016]	[0.018,0.040]	[-0.030,0.010]	[-0.009,-0.002]	[-0.014,0.003]
First-Stage F-Statistic	70	69	29	69	29
Observations	11448	10390	1057	10391	1057

Notes: The Tourism Industry outcome is a binary variable with 1 indicating the household head or principal earner works in a tourism-related industry and 0 indicating the individual works in a non-tourism industry. The outcome for the likelihood of being below the poverty line is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

and 4 occupations, which are managers, professionals, and similar positions, also show no statistically significant effect of tourism on per-capita expenditures, in line with the expectations of the third hypothesis.

In general, these findings suggest partial support for the hypotheses related to the labor force. They are also consistent with other studies. On the one hand, as expected, there are statistically significant effects on the real consumption levels of households in non-tourism service sectors and in the tradable manufacturing sector. On the other hand, the lack of an observable effect for real consumption or poverty among the tourism labor force is at odds with the original hypothesis. This finding is particularly intriguing given the prediction that the greatest benefits would be experienced by tourism workers relative to workers in other sectors of the economy. According to Moretti (2010), the predicted increase in earnings for workers is contingent upon imperfect geographic labor mobility and perfect mobility of labor between sectors. A lack of an effect on tourism sector workers could suggest that tourism workers are in fact relatively immobile between different industries and regions. As a result of data limitations, it is not possible to consider these mechanisms in this study. The finding that skill-level 2 occupations see the greatest increase in consumption is, however, consistent with a high level of sectoral mobility. This finding partially supports the third hypothesis regarding the occupations that would benefit from tourism.

7.3 Heterogeneous Effects Across the Consumption Distribution

Finally, I present results on where in the consumption distribution the benefits of tourism revenues accrue. Hypothesis 4 predicted that the benefits of tourism will accrue primarily to those in the lowest quartile of the per-capita expenditure distribution. The reasoning was that households working in tourism, with its preponderance of skill level 1 and 2 occupations, are disproportionately likely to be in the lowest quartile of the expenditure distribution. This logic will require further attention in light of the findings that Hypothesis 2 could not be proven and that Hypothesis 3 holds for mid-skilled but not low-skilled workers.

I test the fourth hypothesis using an Instrumental Variable Quantile Regression (IVQR). A quantile regression approach allows for a more reliable analysis of differences in effects based on consumption than, for

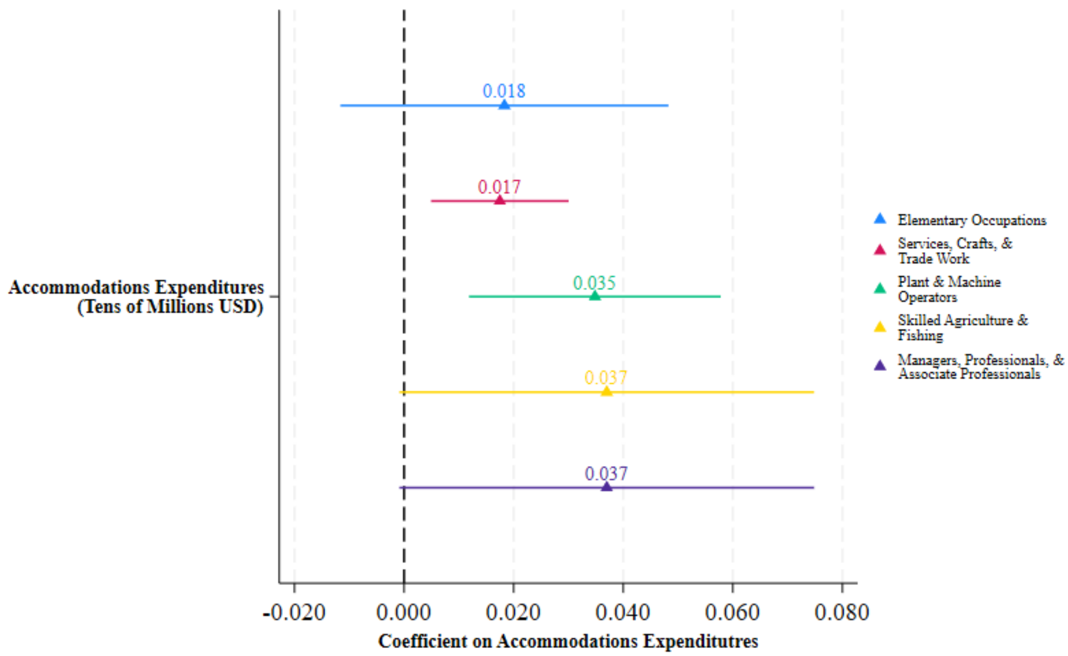


Figure 7: **Cross-Section Comparison of IV Coefficients Across Occupation Categories In Urban Areas**

Shown are the coefficients and 95 percent confidence intervals for the separate regressions of real per-capita household consumption expenditures on development area accommodations revenues for households in different occupations. Accommodations expenditures corrected for inflation to 2024 USD values.

example, separating the sample by quartile and then conducting individual regressions for each quartile. The IVQR analysis for this paper follows the methodology of Kaplan and Sun (2017), using the associated SIVQR STATA package.[33] I once again use the urban sub-sample of the main repeated cross-section household dataset. IVQR uses a minimum distance procedure that relies on a sufficient number of observations in each quantile, conditional on controls and dummy variables. The urban sample is not large enough to run the IV Quantile Regression approach when using the full set of baseline controls and dummy variables. Therefore, for the IV Quantile Regression analysis, I limit the controls to the household size, the sex of the household head, and a binary variable indicating whether or not the year of the observation is earlier than or equal to 2010.

Table 7 presents the results for the analysis. Among the urban sample, the lowest quartile of the expenditure distribution is where all of the statistically significant effects of increases in tourism are concentrated. This aligns with the fourth hypothesis, and with with the previous finding that positive tourism shocks reduced the likelihood of households being below the poverty line. Thus, Hypothesis 4 holds, despite the lack of validation of Hypothesis 2 and the partial validation of Hypothesis 3. The finding that mid-skilled urban workers in non-tourism industries benefit most from tourism suggests that they are the channel through which Hypothesis 4 operates.

We should note that urban households are on average more affluent than rural households (see table 2); therefore, many households in the lowest quartile of the urban distribution are rather high in the national distribution that includes rural households. The upper bound for the lowest quartile of the urban household expenditure distribution is 2157.24 USD, which has the percentile value of 33.55 in the full expenditure distribution.[34]

Although the urban households do consume more on average than rural households, the poorest quartile of urban households are overall still in the bottom third of the national expenditure distribution. Thus, the benefits of tourism primarily accrue to poorest third of Jamaican households. These outcomes align with the goal of tourism growth generating welfare improvements for lower-income Jamaicans.

7.4 Robustness to Alternative Explanations

I now aim to provide plausible explanations for the mechanisms underlying these observed effects. In order to do this, I employ the panel dataset. This does not allow for a one-to-one comparison of the findings because of limited observations, but it still can provide useful insight on the labor force and how it is affected by tourism booms. The panel regression allows me to limit my regressions to households that are in the same industry for each period in which they are observed. In contrast, the cross-sectional results in table 6 reflect the effects of tourism shocks on incumbent workers in an industry, as well as those who are newly hired. The panel regression also allows for comparison of the effects of tourism on households depending on the segment of the distribution in which they fall in the first period they are observed. This can provide a more precise view of where in

[33] Source: Kaplan (2023)

[34] Based on author's calculations from the Jamaica Survey of Living Conditions. See figure 22.

Table 7: IV: Quantile Regression of Log Per-Capita Consumption Expenditure on Tourist Accommodations Expenditures - Urban Households

	Per-Capita Expenditure Quantiles		
	25	50	75
	(1)	(2)	(3)
Tourism Expenditure (Tens of Millions USD)	0.004***	0.001	-0.001
	(0.001)	(0.001)	(0.408)
	[0.002,0.006]	[-0.001,0.004]	[-0.800,0.799]
Observations	11125	11125	11125
Smoothing Bandwidth	95	95	95

Notes: The results above show the coefficient estimates, standard errors in parentheses, and the 95 percent confident intervals in brackets. The above columns represent the coefficient estimates for the $25th, 50th$, and $75th$ quantiles of the Log per-capita expenditure quantiles. Each regression includes a dummy variable indicating whether or not the year is earlier than or equal to 2010. The number of observations in each column is the overall number of urban households in the estimation sample, not the number of households in each quantile. *** p<0.01, ** p<0.05, * p<0.1. All values are provided in 2024 U.S. dollars.

the expenditure distribution households are experiencing the greatest improvements in consumption because of tourism. Unfortunately, because the unbalanced panel has fewer observations than the cross-section in order to maintain statistical power in the first and second stages, I keep both urban and rural households as well as Kingston in this sample. Therefore, this regression does not provide a one-to-one comparison with the main results of the paper, though I argue that its results can still be informative.

Table 8 displays the results of the fixed effects instrumental variable regression of annual real per-capita expenditures on development area tourism revenues, with disaggregations according to selected sectors and the quartiles in which households fall. The first column shows results for the regression using the full sample with all sectors and all quartiles. As expected, on average there is no effect for tourism revenues on household expenditures.

Columns 2-4 show results for households that work in either “Non-Tourism Services” or “Manufacturing” during all periods in which they are observed. Column 2 and column 4 present findings for the 1st and 4th quartiles, respectively, but both have weak first stages and it is therefore not possible to draw strong conclusions from them. Column 3 has a strong first stage and is the subset of households that fall in the 2nd and 3rd expenditure quartiles during the first period in which they are observed. For these households, the coefficient on tourism earnings is 2.5 percent per 10 million USD increase in revenues. This finding is significant at the 0.001 level.

The final three columns are regressions for households that work in tourism for *at least* one of the periods in which they are observed. Once again, the regressions for the 1st and 4th quartiles in the 5th and 7th columns present with weak first-stage results, making interpretation difficult. Column 6 presents findings for households in either the 2nd or 3rd expenditure quartiles in their first period; the coefficient is again positive

Table 8: IV-Panel Per-Capita Consumption by Industry

	All Sectors	Non-Tourism Services & Manufacturing			Ever Employed in Tourism		
	All Quartiles	Quartile: 1	Quartiles: 2-3	Quartile: 4	Quartile: 1	Quartiles: 2-3	Quartile: 4
Tourism Expenditure	-0.003	0.083**	0.025***	-0.030*	0.091**	0.020**	-0.022
(Tens of Millions USD)	(0.007)	(0.038)	(0.008)	(0.015)	(0.041)	(0.009)	(0.018)
	[-0.017,0.011]	[0.006,0.159]	[0.008,0.042]	[-0.061,0.000]	[0.008,0.175]	[0.002,0.038]	[-0.058,0.013]
First-Stage F-Statistic	26	8	64	13	11	59	13
Observations	6032	313	1133	713	225	826	543
Number of Clusters	58	45	53	45	39	46	38

Notes: Non-Tourism Services and Manufacturing observations are those households that are employed in either of those sectors for all periods in which they are observed. Ever Employed in Tourism is a sample of households that are employed in a tourism-related industry for at least one of the periods in which they are observed. Each result above includes the coefficient estimate, followed by the standard error in parentheses. Regressions are separated by the initial quartile in which a household is observed. All regressions include a vector of household fixed effects and year fixed effects. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

and statistically significant. The magnitude is 2 percent. These findings shed further light on the industries in which workers benefit the most from local tourism shocks. They further support the quantile regression results by indicating that it is Jamaican households from the lower middle to the upper middle of the national consumption distribution that are experiencing the greatest benefits from tourism booms. The 6th column leaves open the possibility that there is in fact some impact on workers in the tourism industry, showing that a household with at least one period in the tourism industry does indeed benefit from a tourism shock.[35]

Next, we can consider the concern that the observed results are driven by the inflationary effects of the tourism shock, even when using regional price index adjusted expenditure and observing decreases in the likelihood of a household being in poverty. One prediction from the literature that seemed unlikely to be true in the short-to-medium run in the Jamaican case was the expectation of the localized shock increasing rent on average. In order to formally test for inflationary effects on rent, I regress real per-capita average monthly rent payments on development area tourism revenues for the sample of urban households. This regression uses the main cross-section dataset. I find no effect of tourism earnings on real per-capita rent expenditures for households, as shown in table 9. The strong first Stage F-Statistic of 72 gives further support to the accuracy of the null effect.

To further confirm that the main results are not merely capturing inflationary pressures, I run regressions disaggregated by the employment type of a household. This refers to whether the household works in the private sector, for the government, or as an own-account worker. While the wages of private sector and own-account employees are likely to be affected by short-run changes in market forces in their local communities, government employee salaries are determined by the national budget approved by the Jamaican Parliament for each fiscal year. Therefore, short- to medium-run inflationary effects can be expected to result in decreases in real per-capita expenditures for government employees, whose wages will not respond to a sudden tourism shock. In table 10, I show the results of expenditure regressions disaggregated by the employment type of a household for those living in urban sub-districts. Column 3 shows a lack of any effect of increased tourism earnings on real

[35] There are too few households employed in tourism in all periods to limit the sample to only that segment of the workforce, so it is not possible to precisely determine the effects for households employed only in tourism across all observations.

per-capita expenditures for government workers. Only those working in the private sector are shown to have an increase in real consumption levels as a result of a positive tourism shock. These results lend further support to the prediction of few inflationary effects of tourism on local communities in the Jamaican context.

Table 9: IV: Regression of Real Per-Capita Rent Expenditure on Tourism Revenues

	Real Per-Capita Rent
	(1)
Tourism Expenditure (Tens of Millions USD)	10.620
	(6.772)
	[-3.158,24.397]
First-Stage F-Statistic	72
Observations	11442

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

A lack of significant negative consumption impacts for non-private sector employees suggests a lack of the crowding-out effects observed by Allen et al. (2021) in the Barcelona setting. While this study does not speak to long-term welfare impacts of tourism on Jamaican communities, this does suggest that, at least in the short to medium run, there are no significant welfare losses for segments of the population not involved in the tourism sector.

Table 10: IV: Real Per-Capita Consumption By Employment Type

	Real Log Per-Captia Expenditure		
	Private Sector	Government	Own Account Worker
	(1)	(2)	(3)
Tourism Expenditure (Millions USD)	0.002*	-0.002	0.000
	(0.001)	(0.001)	(0.001)
	[0.000,0.003]	[-0.005,0.000]	[-0.002,0.003]
First-Stage F-Statistic	59	23	32
Observations	10295	2032	11819

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

8 Discussion

This paper investigates the effects of growth in tourism revenues on the consumption and welfare of households in a developing country. There are two principal questions guiding the analysis. First, does an increase in tourist expenditures in a specific area produce an increase in real per-capita consumption for households in that area? Second, to what extent are consumption increases observed among households with lower skills and lower consumption? I combine novel micro-datasets of tourist activity and household expenditure surveys in Jamaica spanning two decades, and spatially link the observations between 2001 and 2021 across consistent spatial units, to produce a comprehensive view of variations in tourism activity and household spending across the study period. I employ a shift-share instrumental variable identification strategy that harnesses two key aspects of tourist behavior: variation in the exposure of different areas of Jamaica to tourists from different regions of origin, and plausibly exogenous variation in the total spending by tourists of different origin groups in Jamaica from year to year. Armed with this identifying variation, this study analyzes the effects of variations in annual tourism revenues on per-capita household expenditures, poverty, and six disaggregated spending categories.

I find strong evidence that larger tourism revenues generate a positive and economically significant effect on real consumption levels for urban residents in the affected areas. These findings are driven by households working in non-tourism services and manufacturing. Among urban households in Jamaica, increases in tourism earnings generate statistically equivalent percentage increases in per-capita food and non-food expenditures, though the dollar value of increases in non-food consumption are larger than those for food consumption. These findings also indicate that households use a sizable share of their increased consumption ability to invest in a key measure of welfare: health and medical services. Combined with meaningful decreases in the likelihood of households falling below the poverty line, I interpret these results as indicating that tourism revenues generate improvements in the welfare of households in areas experiencing tourism growth.

A particularly important result of this paper, related to the welfare of local households and larger discussions surrounding tourism, is the lack of evidence for significant inflationary effects of tourism on local populations. This stands in stark contrast to the experiences of local residents in destinations where urban tourism results in significant inflationary effects (Allen et al. 2021; Almagro and Domínguez-Iino 2025). This is in line with predictions that the "enclave" style tourism of all-inclusive resorts constrains short-run inflationary effects. However, given the data, it is not possible to precisely determine the reasons for the lack of observed changes in cost of living.

The results from this study provide three key insights on the effects of tourism on the Jamaican labor force. The first is that, as predicted by the literature on labor demand shocks for labor-intensive industries, increases in tourism revenues in a given area generate increases in the likelihood that households in that area are employed in the tourism industry.

The second result was not expected. I find that a tourism shock increases real consumption for households working in non-tourism industries. There is a similar story for poverty, with no observable benefit for households employed within the tourism sector. This result may be in part because of a lack of identifying variation in the first stage when limiting the sample to only tourism employees, but it is still notable considering long-running complaints within the tourism sector in Jamaica about low wages and poor working conditions. I provide additional support to these findings by employing an unbalanced panel dataset of households that is a subset of the repeated cross-section utilized for the main analysis. This regression, though not a one-to-one match with the original, reveals that the positive effects of tourism on consumption are felt principally by workers who are employed in either non-tourism services or manufacturing throughout each period they are observed. The panel does, however, reveal statistically significant effects for households that are employed within the tourism industry for at least one period, although lack of statistical power prevents limiting the sample to households exclusively employed in tourism across each observation.

The third labor-related result is that the skill level of occupations and the specific type of occupation are particularly relevant to which parts of the labor force benefit from tourism. Increases in an area's tourism primarily benefit residents working in either "Service, Crafts, and Trades" or "Plant and Machine Operators", both of which are skill-level 2 occupations as defined by the ISOC (International Labour Office 2012). This result is consistent with the result that workers in non-tourism services and manufacturing experience the greatest benefit from tourism growth. Particularly important to the aims of the Jamaican government, and more generally to tourism-based development, is the finding that there are no observed effects for skill-level 1 (low-skilled) occupations, though they represent a relatively large share of the tourism labor force. There are also no observed effects on skilled agricultural workers, despite the aim of improved linkages between the tourism and agricultural sectors. However, tourism can drive improvements in well-being among those in mid-skilled occupations, who collectively represent roughly 75 percent of the labor force (including both urban and rural). This demonstrates the ability of tourism to generate gains that benefit a broad segment of the population. At the same time, the lack of effect for those who are low-skilled, or perform skilled agricultural work, demonstrates the potential limits of tourism's ability to benefit those at the lowest end of the skill distribution. This is a matter for concern because the lowest-skilled workers often are the poorest in a population.

Finally, this paper shows that tourism can produce improvements in consumption and well-being for households at or near the poverty line. This result is tempered by the fact that these effects are only observed for urban households. Results of the quantile regression show that, for urban communities, the positive effects of tourism on earnings are found in the lowest expenditure quartile. However, the real expenditure levels of urban households in the first quartile in fact span the lowest and second-lowest quartiles when considering the full national sample. This is further supported by the panel industry regressions, which show households in the 2nd and 3rd quartiles benefiting from increased tourism revenues. Once again, this result shows both the inclusive potential for tourism services and their limitations. Although a broad range of urban households are shown to benefit, the lack of observed effects for rural sub-districts, where the poorest Jamaicans live (see table 2), shows the difficulty of harnessing back-linkages to generate prosperity across industries and geography.

8.1 Implications For Policy and Research

What are the implications of these results for policy and research into tourism-based development? This paper shows that tourism can be an effective means for generating employment in the context of local communities in a middle-income country such as Jamaica. This suggests that policies meant to encourage tourism investment, such as those in Jamaica providing relief from taxes and tariffs on productive inputs for tourism-sector firms, have the potential to be welfare-improving for communities directly exposed to tourism. The extent of these policies must, however, be weighed against the magnitude of the economic benefits to households, and, importantly, the magnitude of the change in tourism revenues needed to generate real gains in well-being. The scale and growth of the Jamaican tourism industry over the previous two decades has assured that households in communities involved in tourism have experienced some benefit from the sector. This scale has been achieved in part by proactive policies aimed at attracting foreign direct investment in new accommodations. A full accounting of whether the welfare benefits experienced by households are commensurate with the government investment is beyond the scope of this paper and would be an informative area of focus for additional research. That being said, these results do demonstrate the potential for welfare gains that are economically meaningful in a context that is comparable to those in a number of other countries.

This study presents important implications for research and policy related to the types of tourism in which a country and its regions choose to specialize. While there has been substantial criticism of the all-inclusive style of tourism common in many developing country contexts, due to its lack of integration with local communities (Çiftçi et al. 2007; Issa and Jayawardena 2003), this paper suggests that lack of integration may in fact be *beneficial* for local welfare in certain contexts. This paper does not find significant short-run impacts on the cost of living for residents of tourism-exposed development areas. This is particularly notable when we consider that tourists to Jamaica are on average significantly more affluent than most residents in the localities in which they vacation.[36]

This adds an important dimension to the study of the effects of demand shocks on local economies. As Allen et al. (2020) and Almagro and Domínguez-Iino (2025) show, integrated residential tourism can significantly alter living costs and amenities for local residents, with some experiencing considerable losses. This has contributed to local backlash against mass tourism in some settings.[37] However, in the case of enclave-tourism settings, a key aim has been to *increase* the engagement of tourists with the locals of the areas they visit.[38] In Jamaica, protests against the tourism industry have largely related to low wages[39] and poor working conditions, but not to the presence of tourists themselves, as has been observed in places such as Barcelona. Research investigating the trade-offs between the two tourism styles, and the optimal type of tourism for specific contexts, would be a natural extension of this work. For policymakers aiming to develop local tourism industries, it is important to

[36] According to the Ministry of Tourism Exit Surveys, on average 50 percent of tourists have an income over 60,000 USD. See table 40 in Appendix A.2.

[37] In Barcelona for example: https://www.cnn.com/travel/barcelona-reckons-with-overtourism-summer-2025

[38] For example, the Jamaican government has in recent years begun pushing for a greater supply of short-term rental offerings such as AirbnB, which has grown considerably since arriving on the island in 2015.

[39] For example: https://www.jamaicaobserver.com/2024/11/18/bartlett-meeting-tourism-players-another-hotel-stages-protest

consider factors such as the difference in income levels between local residents and the tourists to whom an area is catering, differences in preferences over amenities between the two groups, and the extent to which tourism growth can benefit workers in other sectors via spillovers. The extent to which tourism is truly inclusive and welfare-improving for the majority of a local population will depend crucially on those elements.

Turning to service-led structural change, the results of this study lend some support to optimistic views of the ability of service industries to generate prosperity, while also demonstrating their potential limitations. Tourism does show an ability to absorb lower-skilled labor, as described by Nayyar et al. (2021). However, the results of this study do not make clear the extent to which these new tourism workers benefit relative to workers in other industries. Additionally, the lack of effects for unskilled workers suggests limits on the inclusiveness of certain types of tourism. In terms of spillovers, while we do see benefits to real consumption by manufacturing workers (table 8), it is not clear whether these results are driven by back-linkages to the manufacturing sector (as would be consistent with Faber and Gaubert (2019)), or whether this is primarily driven by a tighter labor market. More broadly, the fact that most beneficiaries from tourism are employed in either non-tourism services or manufacturing bears some similarity to the pattern discussed by Gollin et al. (2016), where natural resource extraction-based economies produce urbanization in "consumption cities," with little industrial production. Whether tourism has the potential to drive growth-enhancing urbanization likely depends on whether tourism linkages with industrial sectors are capable of generating durable growth in Jamaican manufacturing.

A final policy implication is the importance of promoting completion of secondary schooling and up-skilling for segments of a population in order for the spillover effects of tourism services to be felt broadly. Results demonstrate that skill level 2 occupations are consistently the beneficiaries of gains in tourism for residents of an area. Therefore, training programs preparing a workforce for occupations that benefit from the growth of tourism may increase the likelihood that households will reap the benefits of growth in the tourism sector. The skill-level distribution of a country's existing workforce is an important consideration for policy makers evaluating the short-run benefits of policies promoting tourism, particularly the "Sun, Sand, and Sea"-style tourism of Jamaica. If the labor force is largely composed of unskilled or low-skilled workers, it is possible that a smaller share of the population will benefit from tourism spillovers.

There are two additional avenues for future research building on this paper. The first would be to further study the impacts of local tourism shocks, but to also considering internal migration of Jamaican households. While migration from rural areas to tourism zones is a common feature of Jamaican demographic shifts, it is not possible to observe this in the household datasets that I use. The effects observed in this study should be considered as a lower bound on the effects of tourism, as they demonstrate the ability of tourism to raise earnings both for new residents and long-time inhabitants of an area. Fully quantifying the impacts of tourism on economic development will require consideration of the movement of individuals and/or households across regions of Jamaica for work. The second avenue would be to disentangle the extent to which the observed findings are a result of labor-demand-induced increases in earnings, the effects of back-linkages from tourism to other sectors, or the effects of an increase in an area's real wage bill on demand for local goods and services.

Further research considering these short-to-medium run channels could be useful for testing additional policy scenarios and could better capture which residents benefit and to what degree. This would also be useful for understanding the lack of observed impacts on households employed in tourism-related industries, and those who live in rural districts.

8.2 Conclusion

This paper has contributed principally by improving understanding of the magnitude and scope of the effects of tourism services on household consumption and well-being in a developing country. This study has demonstrated that fully characterizing the impact of tourism on local economic development requires considering heterogeneity of households in terms of skill levels and occupations, and that these in turn impact which segments of the consumption distribution benefit from positive tourism shocks. In addition, this study provides a balanced analysis of the potential for tourism to generate economically meaningful gains for major segments of the population, while also qualifying these impacts by showing variations in effects across space and skill level.

Tourism has grown into a major global sector, and presents opportunities for emerging economies to exploit natural and cultural endowments to the benefit of their populations. Tourism services show promise in some instances, but in many cases have been criticized for what is considered their exploitation of local populations. This paper has shown that real economic development gains are possible for significant segments of a local population in the presence of positive tourism shocks, while also showing that tourism-based development involves important trade-offs and limitations. These must be considered in order to best harness its potential for meaningful local prosperity.

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A Appendix

A.1 Instrument Summary Statistics and First Stage Regressions

One concern when using shift-share instrumental variables in a panel setting is the possibility of serial correlation in the shifts. As explained by Borusyak et al. (2024), if shocks g_{kt} are correlated with shocks from period $t-1$, then the estimated effect of tourism on the outcome variables reflects both the contemporaneous effect and the dynamic effect of the previous shock.

Table 11: Summary of Exposure and Shock Measures

	Variable	Mean
1	Mean Exposure Share	0.02
2	Median Exposure Share	0.00
3	Standard Deviation Exposure Share	0.07
4	Mean Shock	0.11
5	Median Shock	0.03
6	Standard Deviation of Shock	0.34
7	Inverse HHI: Shocks	280.19

Notes: These summary statistics cover the SSIV shocks and exposure shares calculated from the Ministry of Tourism Exit Surveys.

A.2 Shift-Share Instrument Diagnostics and Summary Statistics

Table 12: IV: Shift-Share Instrumental Variable Full Summary Statistics

	Year	Mean Exposure	Median Exposure	SD Exposure	Mean Shock	Median Shock	SD Shock	Inverse HHI: Shocks
1	2001	0.02	0.00	0.08	0.05	−0.01	0.29	236.76
2	2002	0.02	0.00	0.08	0.10	0.15	0.26	308.57
3	2003	0.02	0.00	0.08	0.07	0.04	0.11	280.96
4	2004	0.02	0.00	0.07	0.02	0.00	0.20	298.28
5	2005	0.02	0.00	0.08	0.16	−0.02	0.32	267.17
6	2006	0.02	0.00	0.08	0.60	0.25	0.71	267.17
7	2008	0.02	0.00	0.07	0.04	0.10	0.25	270.23
8	2009	0.02	0.00	0.07	−0.04	−0.12	0.33	264.21
9	2010	0.02	0.00	0.07	−0.04	−0.09	0.26	258.78
10	2011	0.02	0.00	0.07	0.02	−0.01	0.11	258.27
11	2012	0.02	0.00	0.07	0.60	−0.13	1.39	250.70
12	2013	0.02	0.00	0.06	−0.10	−0.07	0.14	274.58
13	2014	0.02	0.00	0.07	0.24	0.24	0.14	243.03
14	2016	0.02	0.00	0.07	0.09	0.12	0.22	252.44
15	2017	0.02	0.00	0.06	0.21	0.26	0.29	351.61
16	2018	0.02	0.00	0.06	0.32	0.18	0.44	350.13
17	2019	0.02	0.00	0.06	0.07	0.01	0.21	293.13
18	2021	0.02	0.00	0.07	−0.43	−0.26	0.41	317.39

Notes: Above are summary statistics for the SSIV instruments for all years of the analysis.

I report the AKM standard errors of a shock-level regression following the methodology of Adão et al. (2019) in table 13. In panels A, B, and C, the results of my baseline regression specifications are compared for all Jamaican households, urban households, and rural households, respectively. Adão et al. (2019) propose two methods for calculating the standard errors, and the 95 percent confidence intervals that correct for these biases. I will refer to these calculation approaches as AKM and AKM0, following the terminology in their paper. I show the implied estimates for five different standard error specifications: Homoskedastic, White standard errors, regular clustered standard errors, and finally, the AKM and AKM0 specifications. The AKM approach corrects for potential correlation in shocks between units that have similar exposure shares. For both the AKM and AKM0 standard errors, the results of my baseline regressions remain unchanged. In each of these calculations I consider shock-level variation, and cluster the shocks based on the country.

Table 13: Comparison of AKM Regression Results Across Methodologies

Method	Estimate	Std.Error	P.Value	Left.CI	Right.CI
Panel A: IV Estimates All Regions					
Homoscedastic	0.0113***	0.0043	7.99e-03	0.003	0.0197
EHW	0.0113***	0.0044	9.57e-03	0.0028	0.0199
Reg. Cluster	0.0113	0.0072	1.14e-01	-0.0027	0.0253
AKM	0.0113***	0.0013	0.00e+00	0.0087	0.014
AKM0	0.0113***	0.0015	3.10e-07	0.009	0.015
Panel B: IV Estimates Urban Households					
Homoscedastic	0.0266***	0.0064	7.99e-03	0.014	0.0391
EHW	0.0266***	0.0067	9.57e-03	0.0134	0.0397
Reg. Cluster	0.0266	0.005	1.14e-01	0.0168	0.0363
AKM	0.0266***	0.006	0.00e+00	0.0148	0.0383
AKM0	0.0266***	0.0078	3.10e-07	0.0186	0.049
Panel C: IV Estimates Rural Households					
Homoscedastic	-0.0032	0.0059	0.5856	-0.0147	0.0083
EHW	-0.0032	0.006	0.5917	-0.0149	0.0085
Reg. Cluster	-0.0032	0.0102	0.7531	-0.0232	0.0168
AKM	-0.0032	0.0047	0.4908	-0.0123	0.0059
AKM0	-0.0032	0.0051	0.4478	-0.016	0.0041

Notes: The tourism expenditure is measured in millions and is measured at the level of the development area. *** p<0.01, ** p<0.05, * p<0.1.

Next I show graphs before and after the residualizing of the shift-share tourism arrival shocks.

A.3 Associated Table For Occupational Regression

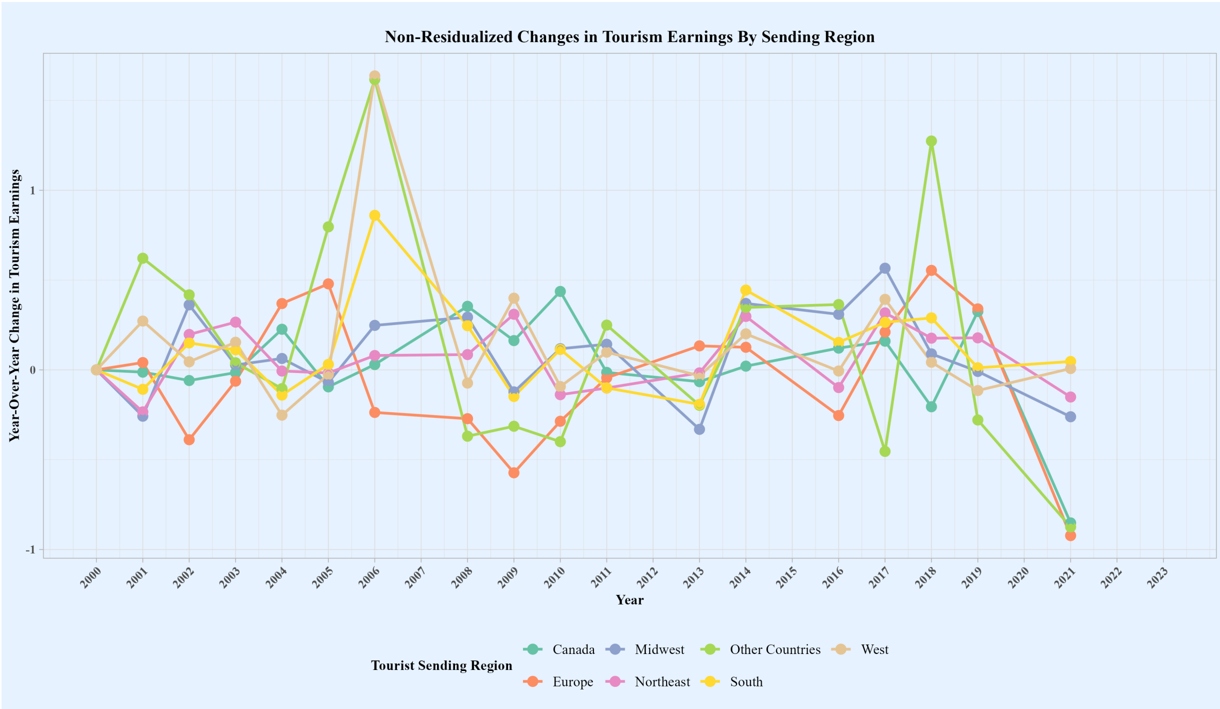


Figure 8: Graphs of Original Shift-Share Shocks

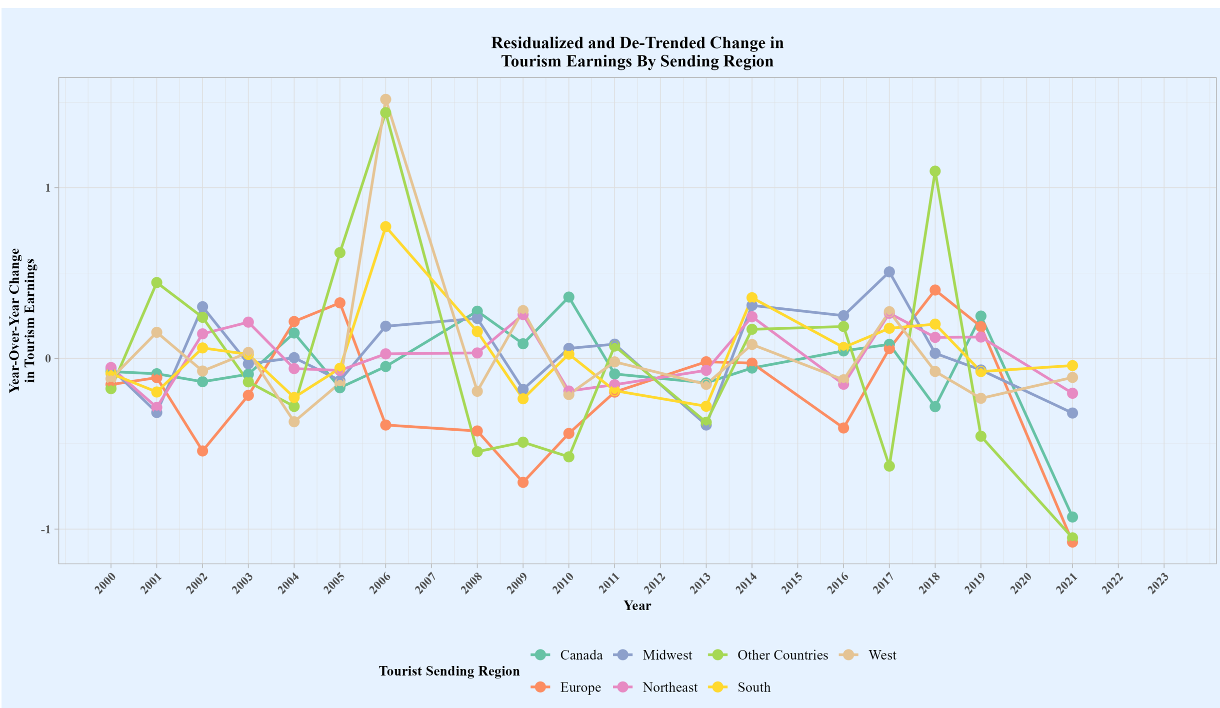


Figure 9: Graph of Residualized Shift-Share Shocks

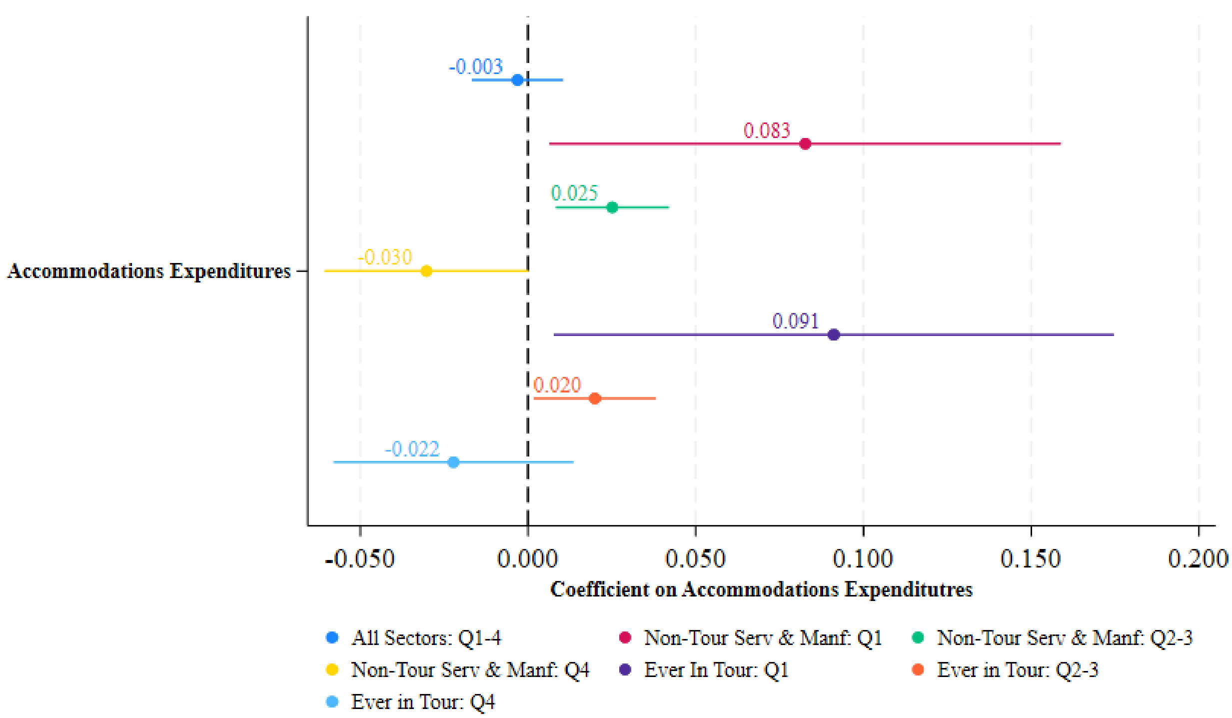
A.4 Panel Regression Results

Table 14: IV: Panel Regression of Baseline Results

	Log Per-Capita Expenditure(USD)		
	(1)	(2)	(3)
Accommodations Expenditures(Tens Millions USD)	-2.8e-03	-2.8e-03	-3.2e-03
	(7.1e-04)	(7.1e-04)	(6.8e-04)
Household Size			-2.1e-01***
			(1.4e-02)
First-Stage F-Statistic	26	26	26
Observations	6032	6032	6032
Standard Deviation	0.711	0.711	0.711
Household Control	No	No	Yes
Household Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes

Notes: Accommodation Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks.

Figure 10: **Panel Comparison of Coefficients Across Industries of Employment**



A.5 All Urban Areas Regression Results

The tables below show the results of regressions for the sample of urban households but not excluding the municipalities of Kingston, Mandeville or May Pen. These localities do not have significant levels of tourism or variation in tourism levels, and thus, the first-stage is considerably weaker when they are included.

Table 15: IV: Relationship Between Tourism Earnings and Log Per-Capita Household Expenditure - All Urban Areas Included

	Annual Log Per-Capita Expenditure		
	Urban Areas (1)	Rural Areas (2)	Rural and Urban (3)
Panel A: Second Stage			
Tourism Expenditure (Tens of Millions USD)	0.001	0.002	0.001
	(0.009)	(0.015)	(0.006)
	[-0.016,0.019]	[-0.028,0.032]	[-0.012,0.013]
Observations	20310	21064	41374
Bootstrapped CI	0.970	0.896	0.915
Number of Clusters	37	60	62
Bootstrapped Hypothesis Test	Yes	No	No
Bootstrapped CI	[.0164, .04148]		
Panel B: First Stage			
Shift-Share Instrument	5.841***	6.071**	5.947***
	(1.358)	(2.738)	(1.748)
	[3.180,8.502]	[0.703,11.439]	[2.520,9.374]
First-Stage F-Statistic	19	5	12
Observations	20310	21064	41374

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 16: IV- Effect of Tourism On Selected Measures of Household Well-Being - All Urban Areas Included

	Log-Expenditure Outcomes			Level Welfare Outcomes		
	Log Per-Capita Food	Log Per-Capita Non-Food	Log Per-Capita Non-Food	Per-Capita Healthcare	Per-Capita Education	Loan Repayment
	(1)	(2)	(3)	(4)	(5)	(6)
Tourism Expenditure (Tens of Millions USD)	-0.006	0.005	0.003	12.075	-3.995	-11.223
	(0.008)	(0.013)	(0.025)	(12.304)	(5.528)	(18.229)
	[-0.023,0.011]	[-0.021,0.030]	[-0.047,0.053]	[-12.879,37.029]	[-15.206,7.216]	[-46.950,24.505]
First-Stage F-Statistic	18	18	20	16	16	16
Observations	20288	20289	12814	18768	18768	12426
Number of Clusters	37	37	37	37	37	37
Bootstrapped CI	[.02488, .02246]	[.01941, .05216]	[.04609, .07081]	[14.26, 51.79]	[14.05, 12.15]	[46.34, 37.87]
Bootstrapped P-Value	0.544	0.781	0.936	0.411	0.589	0.680

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95 percent confidence interval corrected with a wild bootstrap. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 17: IV: Relationship Between Household Poverty and Development Area Tourism Revenues - All Urban Areas Included

	Likelihood of Household Being Below the Poverty Line		
	Urban	Rural	Full Sample
	(1)	(2)	(3)
Tourism Expenditure (Tens of Millions USD)	-0.000	0.004	0.000
	(0.002)	(0.007)	(0.003)
	[-0.005,0.005]	[-0.010,0.019]	[-0.006,0.007]
First-Stage F-Statistic	19	5	12
Observations	20311	21065	41376
Number of Clusters	37	60	62
Bootstrapped Hypothesis Test	Yes	Yes	No
Bootstrapped CI	[.006384, .005309]	[.01159, .0229]	
Bootstrapped P-Value	0.987	0.614	.

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. The outcome is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45, as is the case for the urban sample. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 18: IV - Results By Employment in Tourism-Related Industries - All Urban Households Included

	Employment	Real Log Per-Capita Expenditure		Likelihood of Being The Below Poverty Line	
	Likelihood of Employment in Tourism	Non-Tourism Related Industry	Tourism Related Industry	Non-Tourism Related Industry	Tourism Related Industry
	(1)	(2)	(3)	(4)	(5)
Tourism Expenditure (Tens of Millions USD)	0.003	-0.000	0.016	0.000	-0.004
	(0.003)	(0.010)	(0.011)	(0.002)	(0.004)
	[-0.003,0.009]	[-0.020,0.020]	[-0.006,0.038]	[-0.005,0.005]	[-0.011,0.004]
First-Stage F-Statistic	19	17	9	17	9
Observations	20311	18609	1701	18610	1701
Number of Clusters	37	37	37	37	37
Bootstrapped Hypothesis Test	Yes	Yes	Yes	Yes	Yes
Bootsrapped P-Value	0.517	0.965	0.461	0.859	0.360

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. The outcome for the likelihood of being below the poverty line is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 19: IV - Results By Occupation- All Urban Households Included

	Skill-Level 1 Occupations	Skill-Level 2 Occupations			Skill-Levels 3 & 4 Occupations
	Elementary Occupations	Service, Craft, Trade Work	Plant & Machine Operators	Skilled Agriculture & Fishing	Managers, Professionals & Associate Professionals
Tourism Expenditure (Tens of Millions USD)	-0.003	-0.008	0.006	0.008	0.008
	(0.010)	(0.009)	(0.012)	(0.026)	(0.026)
First-Stage F-Statistic	15	20	14	7	7
Observations	2687	6750	4959	1135	1135
Bootsrapped P-Value	0.751	0.567	0.663	0.744	0.705
Number of Clusters	37	37	37	36	37
Bootstrapped Standard Errors	Yes	Yes	Yes	Yes	Yes
Household Control	Yes	Yes	Yes	Yes	Yes
Household Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. These regressions disaggregate the sample by the occupation category of the household head or primary earner. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 20: IV: Quantile Regression of Log Per-Capita Consumption Expenditure on Tourist Accommodations Expenditures - All Urban Households Included

	Per-Capita Expenditure Quantiles		
	25	50	75
	(1)	(2)	(3)
Tourism Expenditure (Tens of Millions USD)	-0.002	-0.004	-0.005***
	(0.016)	(0.005)	(0.001)
	[-0.033,0.029]	[-0.014,0.005]	[-0.008,-0.003]
Observations	19635	19635	19635
Smoothing Bandwidth	95	95	95

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. The results above show the coefficient estimates, standard errors in parentheses, and the 95 percent confident intervals in brackets. The above columns represent the coefficient estimates for the $25th, 50th$, and $75th$ quantiles of the Log per-capita expenditure quantiles. Each regression includes a dummy variable indicating whether or not the year is earlier than or equal to 2010. The number of observations in each column is the overall number of urban households in the estimation sample, not the number of households in each quantile. *** p<0.01, ** p<0.05, * p<0.1. All values are provided in 2024 U.S. dollars.

Table 21: IV: Regression of Real Per-Capita Rent Expenditure on Tourism Revenues - All Urban Households Included

	Real Per-Capita Rent
	(1)
Tourism Expenditure (Tens of Millions USD)	-4.179
	(6.384)
	[-17.126,8.769]
First-Stage F-Statistic	21
Observations	19619

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 22: IV: Real Per-Capita Consumption By Employment Type - All Urban Households Included

	Real Log Per-Captia Expenditure		
	Private Sector	Government	Own Account Worker
	(1)	(2)	(3)
Tourism Expenditure (Millions USD)	0.000	-0.002	-0.000
	(0.001)	(0.001)	(0.001)
	[-0.002,0.002]	[-0.004,0.000]	[-0.002,0.002]
First-Stage F-Statistic	24	16	15
Observations	8328	1852	5426

Notes: These regressions include the cities of Kingston, Mandeville, and May Pen which are excluded in regressions of the main paper. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

A.6 Rural Household Regression Results

The following tables show the regression results for the sub-sample of rural households. The lack of tourism activity in most rural areas is the main reason for the generally weaker first-stage results for these regressions.

Table 23: IV- Effect of Tourism On Selected Measures of Household Well-Being - Rural Households

	Log-Expenditure Outcomes			Level Welfare Outcomes		
	Log Per-Capita Food (1)	Log Per-Capita Non-Food (2)	Log Per-Capita Non-Food (3)	Per-Capita Healthcare (4)	Per-Capita Education (5)	Loan Repayment (6)
Tourism Expenditure (Tens of Millions USD)	0.005	-0.004	0.023	-4.612	1.873	7.165
	(0.013)	(0.019)	(0.018)	(8.632)	(4.638)	(19.074)
	[-0.021,0.031]	[-0.041,0.033]	[-0.013,0.059]	[-21.886,12.661]	[-7.407,11.153]	[-30.219,44.550]
First-Stage F-Statistic	5	5	6	5	5	5
Observations	21038	21039	13001	19455	19455	12083
Number of Clusters	60	60	60	60	60	60
Bootstrapped CI	[.02036, .03116]	[.04933, .03526]	[.01166, .0558]	[23.57, 12.9]	[8.564, 15.18]	[29.94, 61.3]
Bootstrapped P-Value	0.765	0.806	0.164	0.612	0.628	0.748

Notes: This regression is on the sample of households living in rural enumeration districts. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95 percent confidence interval corrected with a wild bootstrap. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 24: IV - Results By Employment in Tourism-Related Industries - Rural Households

	Employment	Real Log Per-Capita Expenditure		Likelihood of Being The Below Poverty Line	
	Likelihood of Employment in Tourism (1)	Non-Tourism Related Industry (2)	Tourism Related Industry (3)	Non-Tourism Related Industry (4)	Tourism Related Industry (5)
Tourism Expenditure (Tens of Millions USD)	0.004	-0.004	0.018	0.007	0.013
	(0.003)	(0.019)	(0.017)	(0.009)	(0.010)
	[-0.001,0.010]	[-0.043,0.034]	[-0.016,0.053]	[-0.011,0.024]	[-0.008,0.033]
First-Stage F-Statistic	5	5	6	5	6
Observations	20185	18871	1313	18872	1313
Number of Clusters	60	60	59	60	59
Bootstrapped Hypothesis Test	Yes	Yes	Yes	Yes	Yes
Bootsrapped P-Value	0.137	0.780	0.234	0.530	0.245

Notes: This regression is on the sample of households living in rural enumeration districts. The outcome for the likelihood of being below the poverty line is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 25: IV - Results By Occupation- Rural Households

	Skill-Level 1 Occupations	Skill-Level 2 Occupations			Skill-Levels 3 & 4 Occupations
	Elementary Occupations	Service, Craft, Trade Work	Plant & Machine Operators	Skilled Agriculture & Fishing	Managers, Professionals & Associate Professionals
Tourism Expenditure (Tens of Millions USD)	0.020	0.047**	-0.006	-0.071	-0.071
	(0.020)	(0.023)	(0.019)	(0.056)	(0.056)
First-Stage F-Statistic	9	7	4	3	3
Observations	2430	4753	4432	6261	6261
Bootsrapped P-Value	0.332	0.120	0.748	0.335	0.414
Number of Clusters	60	60	60	60	60
Bootstrapped Standard Errors	Yes	Yes	Yes	Yes	Yes
Household Control	Yes	Yes	Yes	Yes	Yes
Household Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Notes: These regressions are on the sample of households living in rural enumeration districts. These regressions disaggregate the sample by the occupation category of the household head or primary earner. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 26: IV: Quantile Regression of Log Per-Capita Consumption Expenditure on Tourist Accommodations Expenditures - Rural Households

	Per-Capita Expenditure Quantiles		
	25	50	75
	(1)	(2)	(3)
Tourism Expenditure (Tens of Millions USD)	0.014***	0.015***	0.010***
	(0.003)	(0.004)	(0.001)
	[0.007,0.020]	[0.008,0.022]	[0.008,0.013]
Observations	20184	20184	20184
Smoothing Bandwidth	95	95	95

Notes: These regressions are on the sample of households living in rural enumeration districts. The results above show the coefficient estimates, standard errors in parentheses, and the 95 percent confident intervals in brackets. The above columns represent the coefficient estimates for the $25th, 50th$, and $75th$ quantiles of the Log per-capita expenditure quantiles. Each regression includes a dummy variable indicating whether or not the year is earlier than or equal to 2010. The number of observations in each column is the overall number of urban households in the estimation sample, not the number of households in each quantile. *** p<0.01, ** p<0.05, * p<0.1. All values are provided in 2024 U.S. dollars.

Table 27: IV: Regression of Real Per-Capita Rent Expenditure on Tourism Revenues - Rural Households

	Real Per-Capita Rent
	(1)
Tourism Expenditure (Tens of Millions USD)	2.153
	(3.578)
	[-5.005,9.312]
First-Stage F-Statistic	5
Observations	20175

Notes: These regressions are on the sample of households living in rural enumeration districts. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 28: IV: Real Per-Capita Consumption By Employment Type - Rural Households

	Real Log Per-Captia Expenditure		
	Private Sector (1)	Government (2)	Own Account Worker (3)
Tourism Expenditure (Millions USD)	0.003	-0.006	-0.003
	(0.002)	(0.004)	(0.003)
	[-0.001,0.008]	[-0.014,0.003]	[-0.008,0.003]
First-Stage F-Statistic	8	7	3
Observations	5901	1229	8958

Notes: These regressions are on the sample of households living in rural enumeration districts. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

A.7 Full-Sample Regression Results

The regressions below include the full-sample of urban and rural households, with no urban areas removed from the analysis. The first-stage for these regressions is generally weaker than the restricted urban sample regressions of the main paper, but stronger than the rural only sample of the prior section.

Table 29: IV- Effect of Tourism On Selected Measures of Household Well-Being - Full Sample

	Log-Expenditure Outcomes			Level Welfare Outcomes		
	Log Per-Capita Food (1)	Log Per-Capita Non-Food (2)	Log Per-Capita Non-Food (3)	Per-Capita Healthcare (4)	Per-Capita Education (5)	Loan Repayment (6)
Tourism Expenditure (Tens of Millions USD)	-0.002	-0.001	0.023	2.368	-2.524	7.165
	(0.007)	(0.008)	(0.018)	(6.649)	(2.954)	(19.074)
	[-0.015,0.011]	[-0.017,0.016]	[-0.013,0.059]	[-10.927,15.662]	[-8.432,3.383]	[-30.219,44.550]
First-Stage F-Statistic	12	12	6	11	11	5
Observations	41326	41328	13001	38223	38223	12083
Number of Clusters	62	62	60	62	62	60
Bootstrapped CI	[.01515, .01556]	[.0167, .02385]	[.01166, .0558]	[10, 20.69]	[8.095, 4.892]	[29.94, 61.3]
Bootstrapped P-Value	0.778	0.856	0.164	0.743	0.597	0.748

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95 percent confidence interval corrected with a wild bootstrap. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 30: IV - Results By Employment in Tourism-Related Industries - Full Sample

	Employment	Real Log Per-Capita Expenditure		Likelihood of Being The Below Poverty Line	
	Likelihood of Employment in Tourism	Non-Tourism Related Industry	Tourism Related Industry	Non-Tourism Related Industry	Tourism Related Industry
	(1)	(2)	(3)	(4)	(5)
Tourism Expenditure (Tens of Millions USD)	0.003	-0.001	0.020*	0.001	-0.003
	(0.002)	(0.007)	(0.012)	(0.003)	(0.005)
	[-0.001,0.007]	[-0.015,0.013]	[-0.004,0.043]	[-0.006,0.007]	[-0.012,0.007]
First-Stage F-Statistic	12	11	9	11	9
Observations	41376	38302	3072	38304	3072
Number of Clusters	62	62	61	62	61
Bootstrapped Hypothesis Test	Yes	Yes	Yes	Yes	Yes
Bootsrapped P-Value	0.255	0.850	0.312	0.768	0.704

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. The outcome for the likelihood of being below the poverty line is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 31: IV - Results By Occupation- Full Sample

	Skill-Level 1 Occupations	Skill-Level 2 Occupations			Skill-Levels 3 & 4 Occupations
	Elementary Occupations	Service, Craft, Trade Work	Plant & Machine Operators	Skilled Agriculture & Fishing	Managers, Professionals & Associate Professionals
Tourism Expenditure (Tens of Millions USD)	-0.001	0.006	-0.006	-0.071	-0.071
	(0.009)	(0.010)	(0.019)	(0.056)	(0.056)
First-Stage F-Statistic	16	15	4	3	3
Observations	5009	11312	4432	6261	6261
Bootsrapped P-Value	0.942	0.713	0.748	0.335	0.414
Number of Clusters	62	62	60	60	60
Bootstrapped Standard Errors	Yes	Yes	Yes	Yes	Yes
Household Control	Yes	Yes	Yes	Yes	Yes
Household Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. These regressions disaggregate the sample by the occupation category of the household head or primary earner. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 32: IV: Quantile Regression of Log Per-Capita Consumption Expenditure on Tourist Accommodations Expenditures - Full Sample

	Per-Capita Expenditure Quantiles		
	25	50	75
	(1)	(2)	(3)
Tourism Expenditure (Tens of Millions USD)	0.013***	0.011***	0.006***
	(0.001)	(0.002)	(0.001)
	[0.010,0.015]	[0.006,0.015]	[0.004,0.008]
Observations	39819	39819	39819
Smoothing Bandwidth	95	95	95

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. The results above show the coefficient estimates, standard errors in parentheses, and the 95 percent confident intervals in brackets. The above columns represent the coefficient estimates for the $25th, 50th$, and $75th$ quantiles of the Log per-capita expenditure quantiles. Each regression includes a dummy variable indicating whether or not the year is earlier than or equal to 2010. The number of observations in each column is the overall number of urban households in the estimation sample, not the number of households in each quantile. *** p<0.01, ** p<0.05, * p<0.1. All values are provided in 2024 U.S. dollars.

Table 33: IV: Regression of Real Per-Capita Rent Expenditure on Tourism Revenues - Full Sample

	Real Per-Capita Rent
	(1)
Tourism Expenditure (Tens of Millions USD)	-5.976
	(5.028)
	[-16.029,4.077]
First-Stage F-Statistic	12
Observations	39794

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

Table 34: IV: Real Per-Capita Consumption By Employment Type - Full Sample

	Real Log Per-Captia Expenditure		
	Private Sector	Government	Own Account Worker
	(1)	(2)	(3)
Tourism Expenditure (Millions USD)	0.001	-0.003*	-0.001
	(0.001)	(0.001)	(0.001)
	[-0.001,0.002]	[-0.005,-0.001]	[-0.003,0.001]
First-Stage F-Statistic	18	14	8
Observations	13957	3010	14217

Notes: These regression results are conducted on the full sample of urban and rural households with no urban areas removed. Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks. *** p<0.01, ** p<0.05, * p<0.1.

A.8 Additional Regression Results

Table 35: IV: Non-Consumption Expenditure

	Urban Log Per-Capita Non-Consumption Expenditure	Rural Log Per-Capita Non-Consumption Expenditure
Tourism Expenditure (Tens of Millions USD)	3.2e-02	3.8e-02
	(5.0e-03)	(2.1e-03)
	[-0.007,0.013]	[-0.000,0.008]
First-Stage F-Statistic	68	10
Observations	7282	11832
Standard Deviation	1.979	1.961
Number of Clusters	39	64
HH Controls	Yes	Yes
DA Dummy	Yes	Yes
Year Dummies	Yes	Yes

*Notes8:*Tourism expenditure is measured in millions and is measured at the level of the development area.

Table 36: IV-Relationship Between Tourism Earnings and Log Household Expenditure By Per-Capita Expenditure Decile Urban Households

	Log Per Capita Expenditure(USD) Separated by Deciles					
	(1-10)	(2-10)	(4-10)	(6-10)	(8-10)	(9-10)
Tourism Expenditure (Tens of Millions USD)	1.7e-02**	1.5e-02*	1.5e-02**	1.2e-02*	1.6e-02*	2.2e-03**
	(5.1e-04)	(5.7e-04)	(5.4e-04)	(5.6e-04)	(6.2e-04)	(6.7e-04)
First-Stage F-Statistic	70	69	54	61	66	48
Observations	11526	11005	9747	8050	5860	4457
Standard Deviation	0.721	0.654	0.589	0.544	0.522	0.535
HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
DA Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Tourism expenditure is measured in millions. All household controls are included.

Table 37: IV - Relationship Between Tourism Earnings and Log Per-Capita Household Expenditure by Rurality

	Urban	Rural
Tourism Expenditure (Tens of Millions USD)	4.3e-03	-8.2e-03
	(8.8e-04)	(1.2e-03)
	[-0.001,0.002]	[-0.003,0.002]
First-Stage F-Statistic	61	18
Observations	17792	18381

Notes: Accommodations Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks. Female Household Head indicates either a single adult female or household with multiple persons for which the household head or principal earner is female.

Table 38: IV: Regression of Log Per-Capita Expenditure By Gender of Individual or Household Head

	Urban		Rural	
	Male-Headed Households	Female-Headed Households	Male-Headed Households	Female-Headed Households
Tourism Expenditure (Tens of Millions USD)	-4.6e-03	1.2e-02	4.8e-03	1.4e-03
	(1.2e-03)	(1.7e-03)	(1.4e-03)	(1.5e-03)
	[-0.003,0.002]	[-0.002,0.005]	[-0.002,0.003]	[-0.003,0.003]
First-Stage F-Statistic	35	20	11	9
Observations	10989	9324	13351	7716
Standard Deviation	0.723	0.725	0.722	0.682
Number of Clusters	45	45	70	70
HH Controls	Yes	Yes	Yes	Yes
DA Dummy	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes

Notes: The tourism expenditure is measured in millions and is measured at the level of the development area.

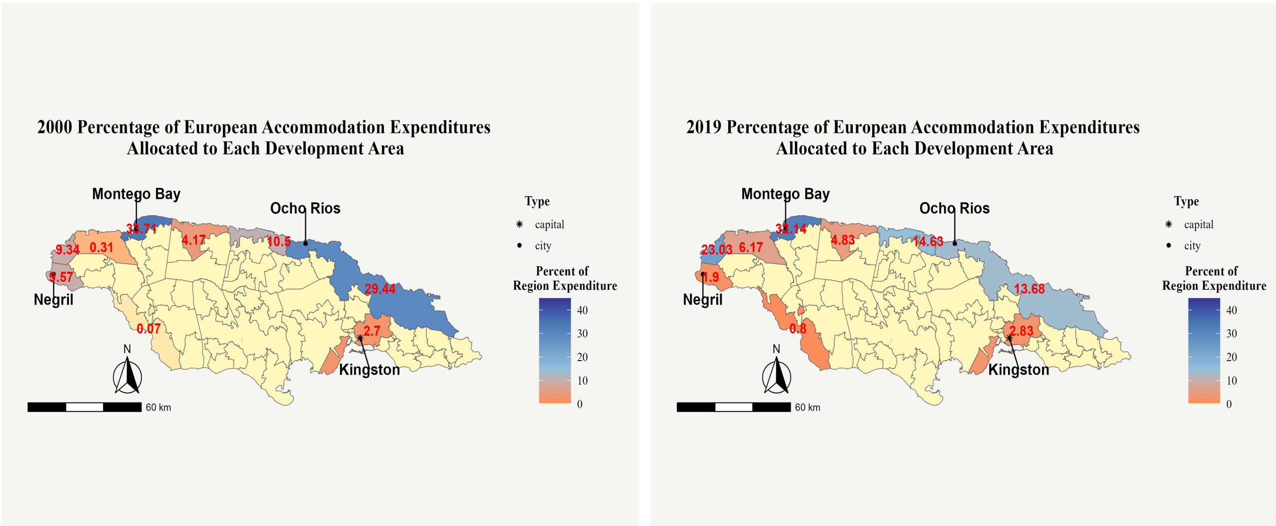
Table 39: IV- Comparing The Impact of Tourism on Urban Homeowners vs. Urban Renters

	Homeowner Expenditure	Renter Expenditure
Tourism Expenditure(Tens of Millions USD)	3.8e-02*	1.1e-03
	(1.4e-03)	(8.1e-04)
	[0.001,0.007]	[-0.002,0.002]
First-Stage F-Statistic	13	15
Observations	6063	9613
Standard Deviation	0.706	0.743
Number of Clusters	34	60
Bootstrapped Standard Errors	Yes	Yes
HH Controls	Yes	Yes
DA Dummy	Yes	Yes
Year Dummies	Yes	Yes

Notes: These regressions exclude the non-tourism urban centers of Kingston, Mandeville, and May Pen. Bootstrapped standard errors are provided in parentheses below the coefficient estimates. The bootstrapped confidence intervals are in brackets below the standard error estimates. Accommodations Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks.

A.9 Tourism Summary Statistics

The graphs below show the geographic distribution of the spending on accommodations by tourists from different sending regions in the years 2000 and 2019. The number in each development area is the share of total accommodations spending by tourists from sending region r allocated to that area in the given year.

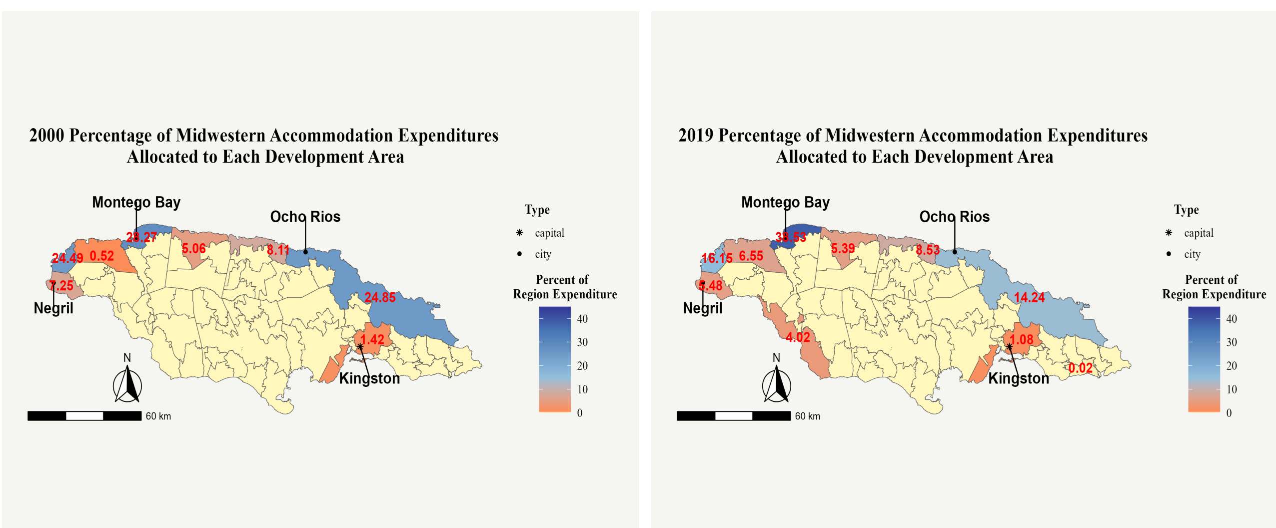


(a) Europeans 2000

(b) Europeans 2019

Figure 11: Variation Over 2 Decades In Spatial Distribution of European Tourist Expenditures

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from Europe between the years 2000 and 2019. The numbers show the share of total European accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Midwestern tourists in that particular development area in a given year.

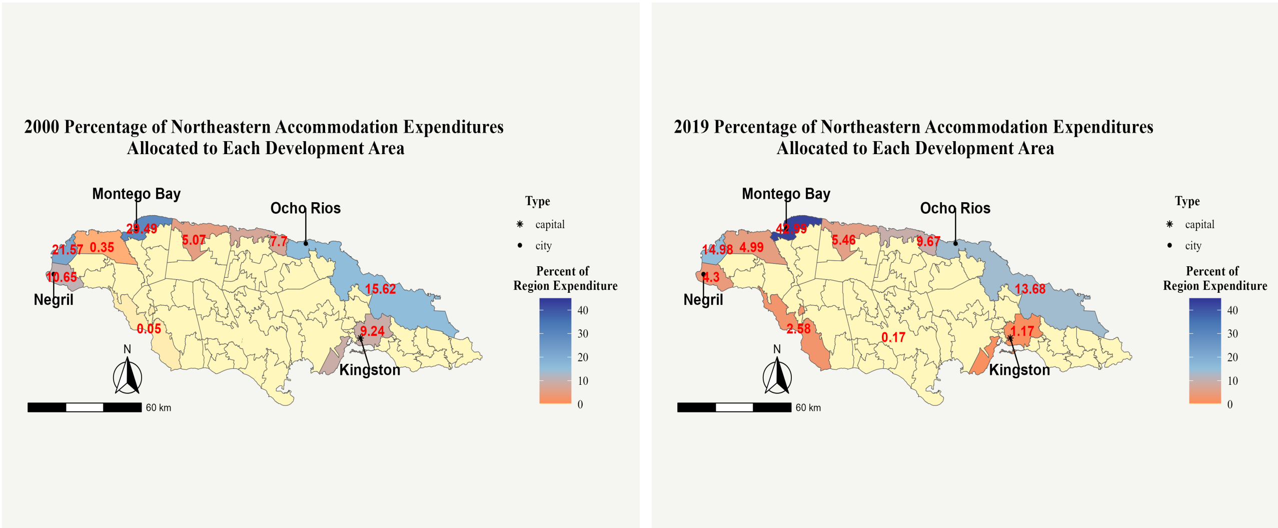


(a) Midwestern Americans 2000

(b) Midwestern Americans 2019

Figure 12: Variation Over 2 Decades In Spatial Distribution of Midwestern American Tourist Expenditures

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from the Midwest of the US between the years 2000 and 2019. The numbers show the share of total Midwestern American accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Midwestern tourists in that particular development area in a given year.

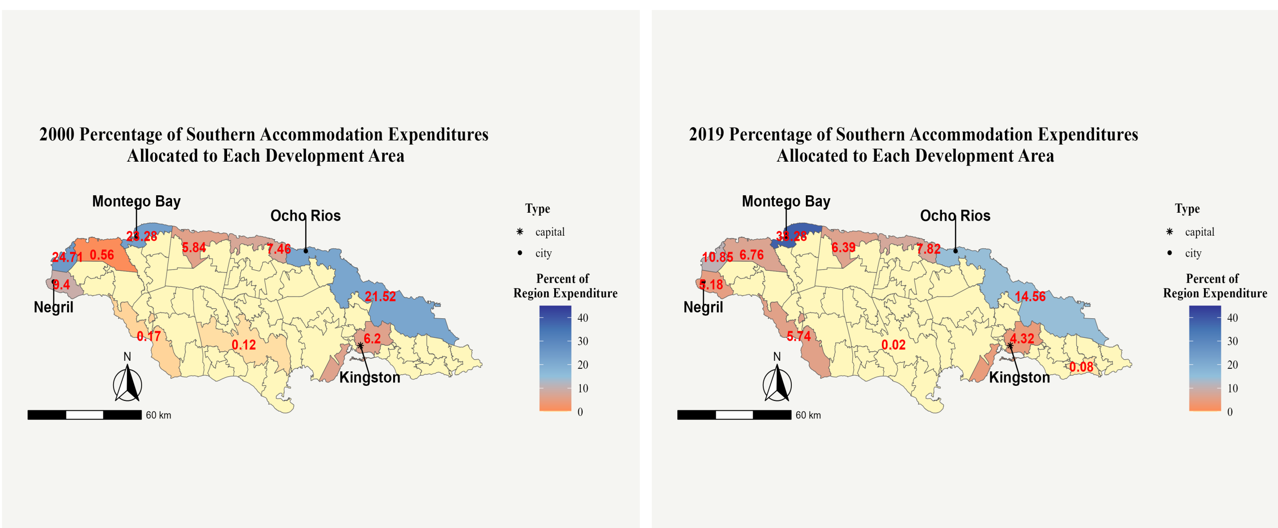


(a) Northeastern Americans 2000

(b) Northeastern Americans 2019

Figure 13: Variation Over 2 Decades In Spatial Distribution of Northeastern American Tourist Expenditures

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from the Northeast of the U.S. between the years 2000 and 2019. The numbers show the share of total Northeastern American accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Northeastern tourists in that particular development area in a given year.

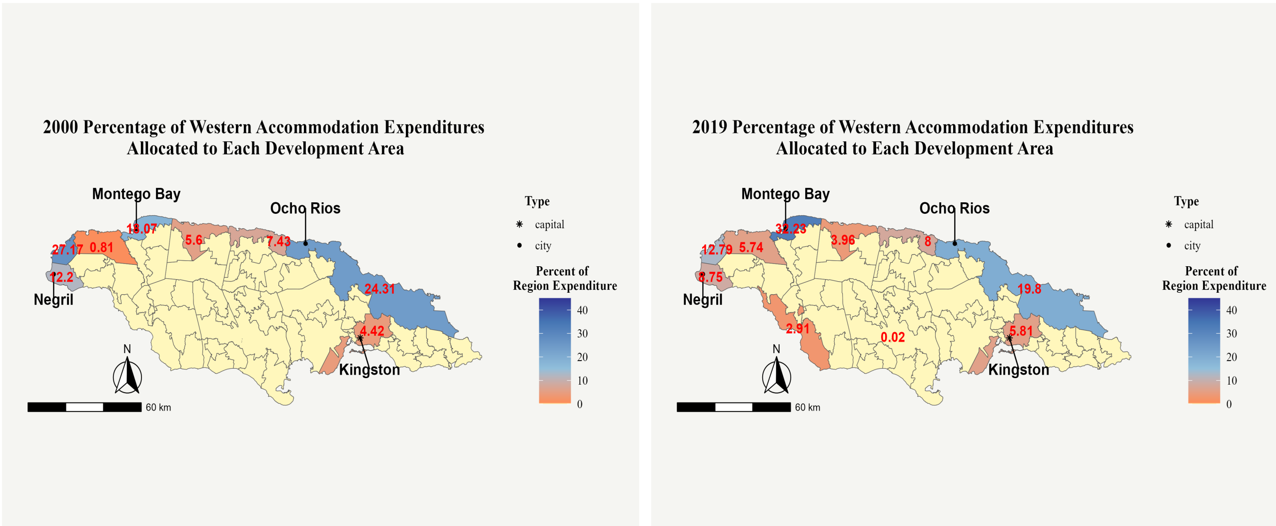


(a) Southeast US 2000

(b) Southeast US 2019

Figure 14: Variation Over 2 Decades In Spatial Distribution of Tourist Expenditures by Visitors from the Southeast of the US

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from the Southeast of the US between the years 2000 and 2019. The numbers show the share of total Southeast US visitor accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Southeastern tourists in that particular development area in a given year.

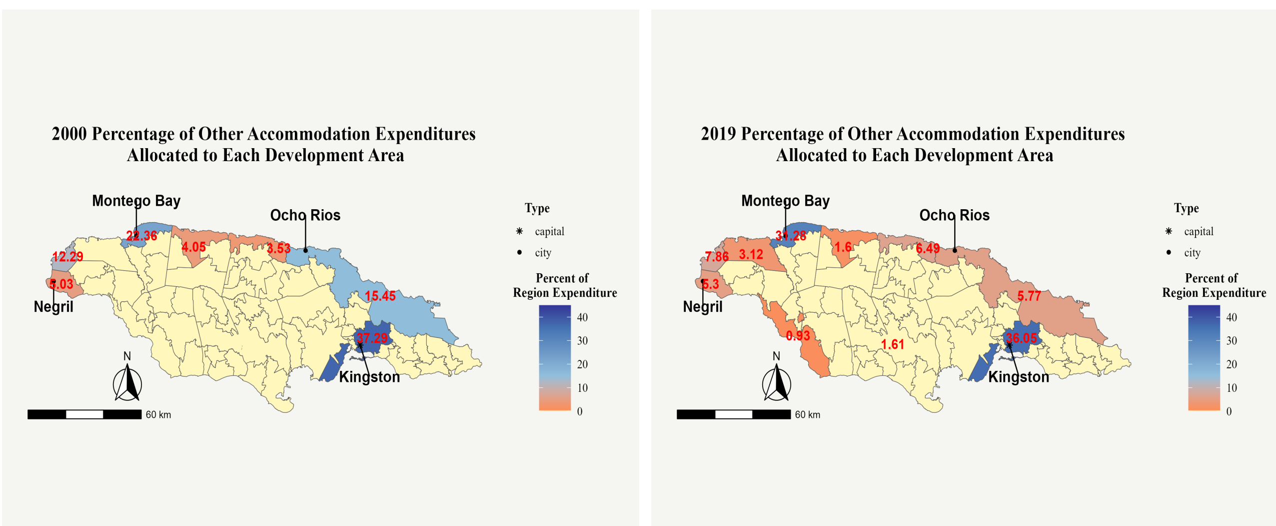


(a) Western US 2000

(b) Western US 2019

Figure 15: Variation Over 2 Decades In Spatial Distribution of Tourist Expenditures by Visitors from the West of the US

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from the West of the US between the years 2000 and 2019. The numbers show the share of total Western US visitor accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Western US tourists in that particular development area in a given year.



(a) Other Countries 2000

(b) Other Countries 2019

Figure 16: Variation Over 2 Decades In Spatial Distribution of Tourist Expenditures by Visitors from Other Countries

Notes: Data comes from the Ministry of Tourism Exit Surveys. This graph shows the changes in the relative popularity of different Jamaican development areas with tourists from all regions other than those currently mentioned including Latin America and the Caribbean and Asia between the years 2000 and 2019. The numbers show the share of total Other Country visitor accommodations spending allocated to particular areas of Jamaica in different years. Yellow shading indicates there was no expenditure by Other Country tourists in that particular development area in a given year.

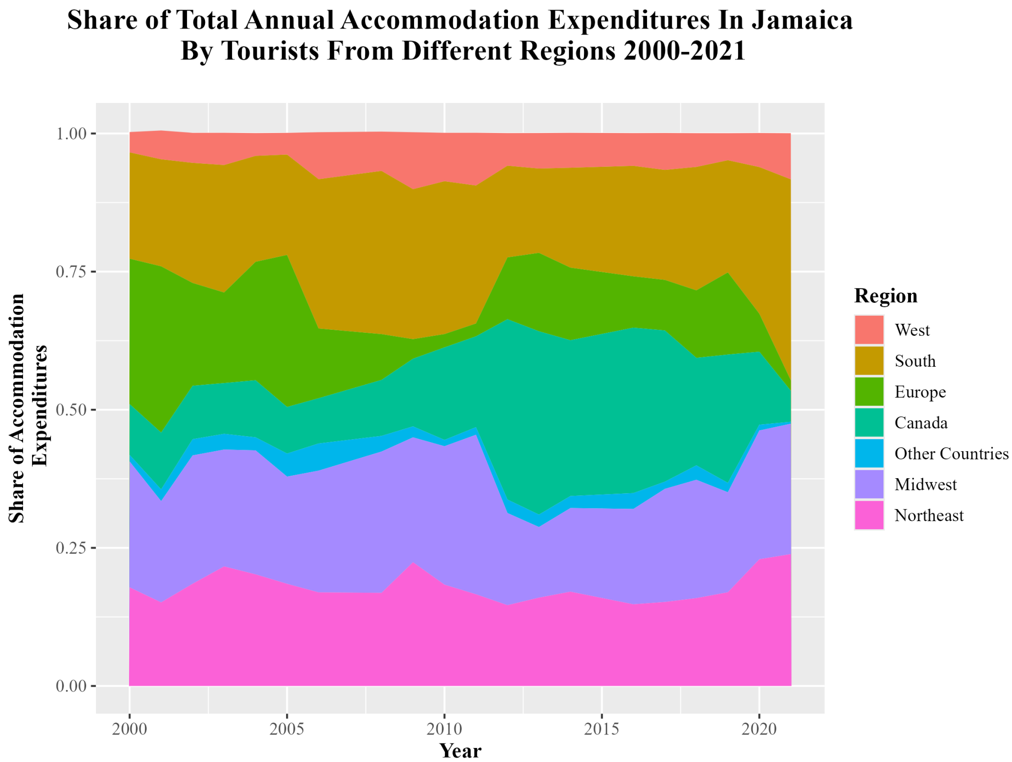


Figure 17: **Share of Expenditures by Region**

Table 40: Tourist Summary Statistics

	Mean	Standard Deviation	Median
Avg. Accom Price per Person	590.03	495.58	520.69
Number of People in Party	1.95	0.96	2.00
Total Cost of Trip	2706.91	1845.23	2350.90
Length of Stay	7.47	5.88	7.00
Visit for Vacation	0.74	0.44	1.00
Return Visitor	0.47	0.50	0.00
Summer Visitor	0.57	0.49	1.00
Income Over US60,000	0.50	0.50	0.00
Observations	78774		

Source: Author's own calculations based on Ministry of Tourism Exit Surveys (20002023).

Table 41: Variation In Tourist Expenditure Levels Over Time Lags

Statistic	1 Year Lagged Value	5 Year Lagged Value	10 Year Lagged Value
Mean Expenditure Change	21.09	46.04	63.53
Median Expenditure Change	7.92	31.06	36.73
Min Expenditure Change	0	0.02	0.02
Max Expenditure Change	235.54	430.92	449.26
SD Expenditure Change	33.35	63.34	89.45

Notes: All expenditures are in millions of US Dollars corrected for inflation to the year 2024.

A.10 Additional Maps And Geographic Summary Statistics

Figure 18: Map of Jamaican Parishes

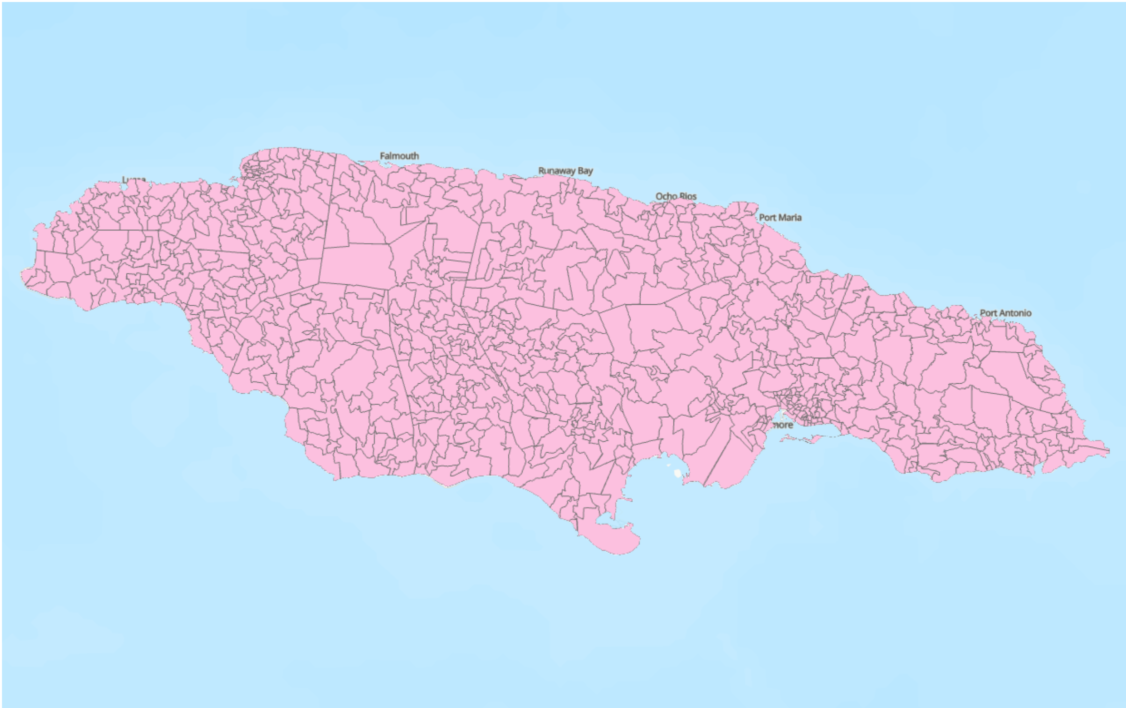


Figure 19: **Jamaica Community Boundaries 2011 (STATIN Geographic Services Unit)**

Table 42: Development Area Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Population	84	32,107.25	37,109.81	1,329	192,044
Area	84	130.52	88.58	3.56	446.33
Population.Density	84	610.98	1,429.67	42.12	8,410.05

Notes: Area is measured in square kilometers. Population is based on the 2011 Census.

A.11 Additional Household and Labor Force Statistics

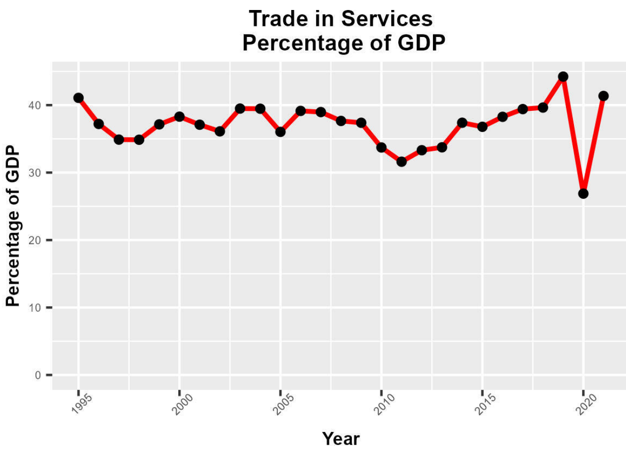


Figure 20: Services Sector Share
Source: World Bank

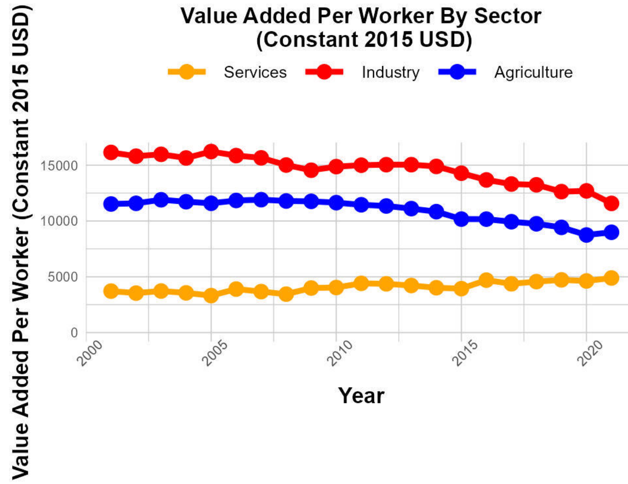


Figure 21: Multi-Sector Value Added
Source: World Bank

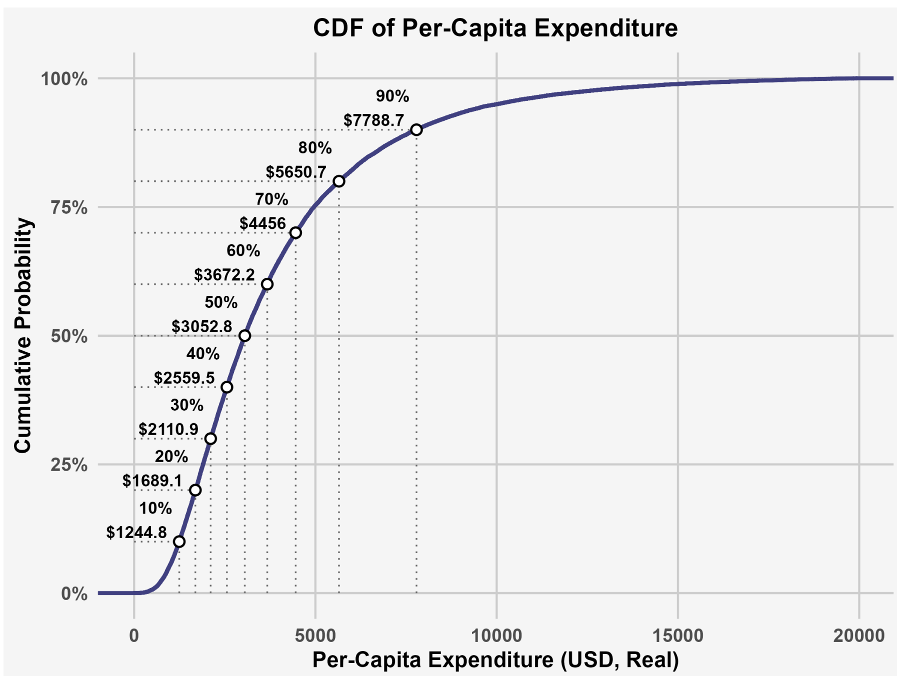


Figure 22: Real Per-Capita Expenditure Distribution of Jamaica Households
Source: Jamaica Survey of Living Conditions

Panel A: Agriculture vs Tourism Sector

Table 44: Agriculture vs. Tourism Services Comparison

	Agriculture		Tourism Services		Difference
	Mean	SD	Mean	SD	T-Stat
Per-Capita Consumption	2965.58	2372.50	3548.34	3046.32	-19.48***
Per-Capita Total Expenditure	3186.53	2767.628	3915.44	3878.217	-18.82***
Per-Capita Food Expenditure	1646.33	1346.511	1655.67	1295.775	-13.87***
Per-Capita Non-Food Expenditure	1319.75	1346.134	1893.32	2194.854	-18.72***
Per Capita Non Consumption Expenditure	341.91	897.240	449.24	1523.801	-8.50***
Non-Food Share of Consumption Expenditure	0.43	0.143	0.51	0.136	-22.66***
Non-Food Share of Tot. Expenditure	0.42	0.140	0.48	0.132	-17.70***
Consumption Share of Tot. Expenditure	0.96	0.083	0.95	0.091	11.81***
Years of Schooling	10.27	3.765	12.86	4.432	-17.95***
Household Decile	5.25	2.818	6.04	2.680	-27.06***
Male HH Head	0.54	0.499	0.61	0.488	0.23
Female HH Head	0.15	0.352	0.34	0.474	-15.37***
Single Male	0.30	0.458	0.04	0.196	18.22***
Single Female	0.02	0.141	0.01	0.096	-3.39***
Observations	6621		2862		9483

Panel B: Manufacturing vs. Tourism

Table 45: Manufacturing vs. Tourism Services Comparison

	Manufacturing		Tourism Services		Difference
	Mean	SD	Mean	SD	T-Stat
Per-Capita Consumption	3940.06	3235.83	3548.34	3046.32	-5.54***
Per-Capita Total Expenditure	4422.46	4229.719	3915.44	3878.217	-5.01***
Per-Capita Food Expenditure	1902.13	1430.458	1655.67	1295.775	-5.07***
Per-Capita Non-Food Expenditure	2037.93	2244.189	1893.32	2194.854	-4.71***
Per Capita Non Consumption Expenditure	627.78	1744.541	449.24	1523.801	-1.48
Non-Food Share of Consumption Expenditure	0.50	0.143	0.51	0.136	-2.40*
Non-Food Share of Tot. Expenditure	0.46	0.138	0.48	0.132	-1.94
Consumption Share of Tot. Expenditure	0.94	0.100	0.95	0.091	1.25
Years of Schooling	12.35	2.790	12.86	4.432	-2.69**
Household Decile	6.43	2.732	6.04	2.680	-5.27***
Male HH Head	0.52	0.500	0.61	0.488	-0.80
Female HH Head	0.25	0.433	0.34	0.474	-2.83**
Single Male	0.20	0.403	0.04	0.196	4.91***
Single Female	0.02	0.156	0.01	0.096	-1.55
Observations	1330		2862		4192

Notes: All statistics are weighted by household size. Panel A reports means, medians, and standard deviations for the full sample. Panel B compares urban and rural households using t-tests with unequal variances. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

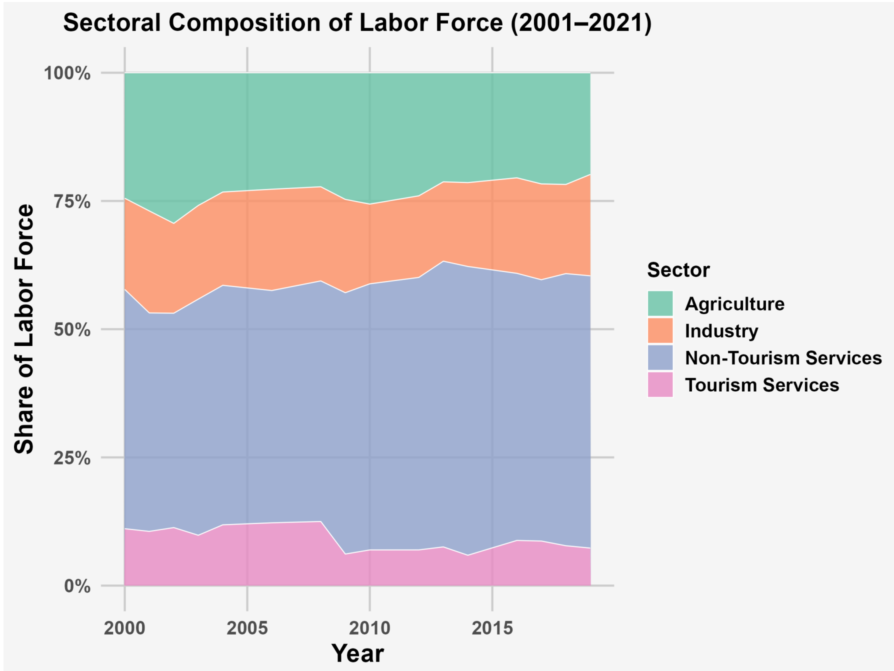


Figure 23: Sector Shares of Jamaican Labor Force
Source: Jamaica Survey of Living Conditions